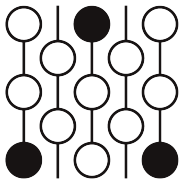
Review of Problem 9 – Humanize



First Proof Third Batch
Reviewer 1
October 8–9, 2026

Not on your report list: read for context; comments welcome.

1 Review

This proposed solution claims partial progress to the general problem, but unfortunately even that is incorrect. Much of the paper begins by citing some of the landmark results of Teichmüller dynamics (e.g. Eskin–Mirzakhani–Mohammadi) without even mentioning the more recent and extremely relevant work of Arana-Herrera–Wright and Benirschke–Dozier–Rached. The stated inputs coming from hallucinated work of Mukamel and McMullen on existence/non-existence of totally geodesic surfaces are also completely incorrect. They AI could not even cite Mirzakhani–Wright correctly (her name is replaced with N. Mazor). Overall, the lack of insight and correctness, as well as the sheer amount of AI slop and hallucinations makes this writeup abysmal.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: -1 points

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: -1 points

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: -1 points

2 Submission with inline comments

3 Statement and setup

Let $\mathcal{M}_g$ be the moduli space of genus $g \geq 2$ Riemann surfaces, $\overline{\mathcal{M}}_g = \mathcal{M}_g \sqcup \partial\mathcal{M}_g$ its Deligne–Mumford compactification, and Δ_G the boundary stratum of stable curves with dual graph G. We are given:

(I) a smooth complex subvariety $V_0 \subset \mathcal{M}_g$ of complex dimension 2, totally geodesic for the Teichmüller metric;

(II) a holomorphic deformation $\mathcal{V} \subset \mathcal{M}_g \times \Delta$ of V_0 over the unit disc $\Delta \subset \mathbb{C}$: a complex submanifold such that, writing $\phi : \mathcal{V} \to \Delta$ and $\psi : \mathcal{V} \to \mathcal{M}_g$ for the projections, each $V_t = \psi(\phi^{-1}(t))$ is a smooth complex surface, and the family is smoothly trivial;

(III) the technical hypotheses (i) (each $\overline{V}_t \cap \Delta_G$ is a smoothly trivial family) and (ii) (compatibility of boundary families across degenerations of strata), all closures being taken in $\overline{\mathcal{M}}_g$.

Fix a stable dual graph Γ with

$$\Delta_\Gamma \;=\; \mathcal{M}_{g_1,n_1} \times \cdots \times \mathcal{M}_{g_k,n_k}, \qquad \pi_i : \Delta_\Gamma \longrightarrow \mathcal{M}_{g_i,n_i}$$

the projection to the i-th factor, and set $\partial V_t := \overline{V}_t \cap \Delta_\Gamma$. We assume $\dim_{\mathbb{C}} \partial V_t = 1$ and that for some index i,

$$\pi_i(\partial V_t) = \{Y_t\} \quad \text{for all } t \in \Delta.$$

Theorem 3.1 (Reduction theorem)**.** *Under these hypotheses, $\pi_i(\partial V_t) = \{Y_0\}$ for every $t \in \Delta$: the projection to the i-th factor is constant in t.* This document proves the implication

$$\big(\textit{hypotheses } (i), (ii), \dim \partial V_t = 1, \pi_i(\partial V_t) = \{Y_t\}\big) + \text{ Input (Mem)} \implies \pi_i(\partial V_t) = \{Y_0\},$$

where Input (Mem) *is the single analytic statement isolated on p. 12 (membership of the family of limiting stable differentials in the affine manifold generated by the V_0 boundary piece at the same face and slope); every other implication is proved in full. The exceptional low-complexity factor cases where Input (Mem) is not needed are proved unconditionally in Appendix A.*

3.1 Notation and reductions

- We work throughout on suitable finite covers: $\mathcal{M}_{g,n}$ is a complex orbifold; by the Kerckhoff solution of the Nielsen realization problem every point has a neighbourhood over which orbifold charts factor through a finite global cover (obtained by fixing a level structure, e.g. a Torelli level ≥ 3, and at the boundary additionally normalizing the plumbing coordinates $T_j \mapsto T_j^\ell$ to kill gluing roots and isotropy, as in standard treatments of the plumbing coordinate system). All convergence, analyticity (Bishop, Rado), and algebraicity arguments below take place on such covers, whose existence makes the projections π_i genuine maps of complex manifolds; the conclusions descend to the coarse spaces and to orbifold quotients by naturality of all constructions. We suppress the covers from the notation.

- $\mathcal{Q}_{g,n} \to \mathcal{M}_{g,n}$ denotes the bundle of (orbifold) pairs (X, q) with q a nonzero meromorphic quadratic differential on X, at worst double poles at the marked points, up to the $\mathbb{C}^*$-action $c \cdot (X, q) = (X, c^2 q)$; $\mathcal{Q}(\mu)$ a stratum prescribed by orders μ of zeros and poles. A stratum

carries its *period coordinate* atlas: after passing to the canonical double cover $\pi : \widehat{X} \to X$ with $\pi^*q = \omega^2$, the map $(X, q) \mapsto [\gamma \mapsto \int_\gamma \omega] \in \widehat{H}^1$ (the anti-invariant cohomology relative to $\Sigma = \{\text{zeros and marked points}\}$) is a local biholomorphism to a complex vector space; strata are thereby complex affine manifolds on which $\mathrm{GL}_2^+(\mathbb{R})$ acts.

- $\Phi : \mathcal{Q}_{g,n} \dashrightarrow \mathcal{M}_{g,n}$ is the forgetful projection (a $\mathbb{C}^*$-quotient on its domain of definition).
- For t near 0 we abbreviate $C_t := \partial V_t$, a complex 1-dimensional closed analytic subset of Δ_Γ (a finite union of curves; by hypothesis (i) it varies in a smoothly trivial family).

3.2 Strategy

Theorem 3.1 combines four inputs:

(a) the *magic wand* theorem of Eskin–Mirzakhani–Mohammadi together with Filip's algebraicity theorem (Theorem 4.1): the closure of any $\mathrm{PSL}_2(\mathbb{R})$-invariant set in a stratum of quadratic differentials is an affine invariant submanifold, algebraic over $\bar{\mathbb{Q}}$ in the moduli projection;

(b) rigidity inputs around Mukamel's theorem (Theorem 4.3): the structure of ΩV for a totally geodesic surface, the non-existence of closed totally geodesic surfaces in high-complexity moduli factors, and the Veech dichotomy (cylinder-only degeneration of lattice surfaces);

(c) the Deligne–Mumford / Mumford–Deligne rescaling theory of Mazor–Wright and of Bainbridge–Chen–Gendron–Grushevsky–Möller (Theorem 4.7), producing affine-invariant degeneration limits ("central Bers sets") and analytic regeneration across nodes with prescribed constant residues (Proposition 4.9);

(d) a final globalization via the identity theorem applied to the holomorphic map $t \mapsto Y_t$ (Lemma 5.3).

In outline: constant collapsed-node combinatorics and slopes (Lemma 5.1, from hypothesis (ii)) make the limiting stable differentials (O_t, q_t) a well-defined 1-parameter family with constant residues (Proposition 5.8); the V_0-anchored membership lemma (Lemma 5.6, which isolates the single external analytic input, Input (Mem)) places this family in the boundary affine manifold generated from ΩV_0; constant residues then allow MD-plumbing regeneration of a holomorphic 2-parameter slice whose boundary disc has nonzero π_i-derivative; the boundary disc lies in $\overline{V_0} \cap \{T_1 = 0\}$, contained in $\overline{C_0}$ by hypothesis (ii) (with the $t = 0$ premise verified in Lemma 6.2), forcing its π_i-image to be the single point Y_{t_*} (Lemma 6.4) — contradicting its nonzero derivative (Proposition 6.3). Hence $t \mapsto Y_t$ is locally constant near 0, and the identity theorem gives $Y_t = Y_0$ for all $t \in \Delta$.

> **Reviewer 1**
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> The argument sketched here doesn't clearly outline what the relevant features of the cited results are or how these big theorems serve the goals of the main problem. It is clear from later mistakes that this work is just wrong.

4 Tools from Teichmüller dynamics

4.1 Teichmüller discs and the totally geodesic condition

A *Teichmüller disc* is the image of a holomorphic totally geodesic embedding of the hyperbolic plane $\mathbb{H} \to \mathrm{Teich}_{g,n} \to \mathcal{M}_{g,n}$ generated by a quadratic differential (X, q): concretely the orbit

$\mathrm{PSL}_2(\mathbb{R})\cdot(X,q)$ projects to $\mathcal{M}_{g,n}$ through the $\mathbb{C}^*$-quotient. A smooth complex subvariety $V \subset \mathcal{M}_{g,n}$ is *totally geodesic* iff every Teichmüller disc tangent to V is contained in V; thus

$$\Omega V := \{(X,q) \in \mathcal{Q}_{g,n} : \text{the disc generated by } q \text{ is tangent to } V \text{ at } X\}$$

is a closed, $\mathrm{PSL}_2(\mathbb{R})$- and $\mathbb{C}^*$-invariant complex submanifold of dimension $2\dim_{\mathbb{C}} V - 1$, and $V = \Phi(\Omega V)$. (The bundle ΩV is the restriction of $\mathcal{Q}_{g,n}$ to V cut down to the $(\dim V - 1)$-plane field of disc tangents along V; its smoothness is classical (the totally geodesic condition identifies TV with an $\mathbb{R}$-linear subspace of the period-coordinate model, cf. [4]), and its $\mathrm{GL}_2^+(\mathbb{R})$-stability follows because the group action takes Teichmüller discs to Teichmüller discs. Cf. also Theorem 4.1(1).)

4.2 Affine invariant manifolds: the magic wand

Theorem 4.1 (Eskin–Mirzakhani–Mohammadi [1], [2]; Filip [3])**.** *Let $\mathcal{Q}(\mu)$ be a stratum of quadratic differentials.*

(1) Every closed $\mathrm{GL}_2^+(\mathbb{R})$-invariant subset of $\mathcal{Q}(\mu)$ is a finite union of affine invariant submanifolds*: immersed submanifolds given locally in period coordinates by linear equations with real coefficients, locally constant along leaves. In particular every $\mathrm{PSL}_2(\mathbb{R})$-orbit closure is an affine invariant submanifold.*

(2) Every $\mathrm{PSL}_2(\mathbb{R})$-invariant probability measure on a stratum is the affine Lebesgue measure on such a submanifold.

(3) (Filip) *Every affine invariant submanifold M projects to a complex algebraic subvariety $\Phi(M) \subset \mathcal{M}_{g,n}$; M itself is quasi-projective in partial compactification and defined over $\bar{\mathbb{Q}}$; the projection of a closed invariant set is closed in $\mathcal{M}_{g,n}$.*

(4) If M is such a submanifold which is $\mathbb{C}^*$-invariant (i.e. stable under $(X,q) \mapsto (X,c^2q)$), then $\Phi(M)$ is closed and stable under taking $\mathrm{PSL}_2(\mathbb{R})$-closures of its points: every fibre of $\Phi|_M$ contains the $\mathbb{C}^*$-orbit $\{(X,c^2q)\}$, so

$$\dim_{\mathbb{C}} \Phi(M) \;\le\; \dim_{\mathbb{C}} M - 1, \qquad \textit{and} \qquad \dim_{\mathbb{C}} \Phi(M) < \dim_{\mathbb{C}} M.$$

Remark 4.2. (a) Statements (1)–(3) are the main theorems of [1], [2] and [3]. (b) In (4) one has equality $\dim_{\mathbb{C}} \Phi(M) = \dim_{\mathbb{C}} M - 1$ exactly when the generic fibre of $\Phi|_M$ is the $\mathbb{C}^*$-orbit alone; the fibre is always at least this. Note the following fundamental example in our setting: for the totally geodesic *surface* $V_0 \subset \mathcal{M}_g$ of our problem, ΩV_0 is a closed $\mathbb{C}^*$- and $\mathrm{GL}_2^+(\mathbb{R})$-invariant submanifold (hence, by (1), affine invariant) of complex dimension 3, and $\Phi(\Omega V_0) = V_0$ has complex dimension $2 = 3 - 1$: here the generic fibre of $\Phi|_{\Omega V_0}$ has complex dimension 1, so ΩV_0 is the total space of a disc bundle over the Riemann surface uniformizing the $\mathrm{PSL}_2(\mathbb{R})$-action; that is, over each point of V_0 the fibre of Φ in ΩV_0 is the 1-dimensional $\mathbb{C}^*$-family of differentials generating the tangent Teichmüller disc (the fibre dimension is 1 for this surface case, and 2 when the base is a curve, as for a Teichmüller curve). (c) A *Teichmüller curve* is, by definition, the projection $\Phi(M)$ of a closed $\mathbb{C}^*$- and $\mathrm{GL}_2^+(\mathbb{R})$-invariant M of complex dimension 3 whose projection has dimension 1; equivalently, the orbit of a quadratic differential with a lattice Veech group. It is a complex 1-dimensional algebraic curve in $\mathcal{M}_{g,n}$, totally geodesically immersed (a union of Teichmüller discs).

4.3 Rigidity of totally geodesic surfaces

Theorem 4.3 (Rigidity inputs [4], [5], [6])**.** *(1)* (Structure of ΩV for a totally geodesic surface) *If $V \subset \mathcal{M}_{g,n}$ is a smooth, closed-in-the-analytic-topology, complex 2-dimensional totally geodesic subvariety, then $\Omega V \subset \mathcal{Q}_{g,n}$ is a closed, $\mathbb{C}^*$- and $\mathrm{GL}_2^+(\mathbb{R})$-invariant complex 3-manifold, affine invariant in the sense of Theorem 4.1(1), with $\dim_{\mathbb{C}} \Phi(\Omega V) = 2$: at a generic point, the tangent space $T\Omega V$ is the (complex 3-dimensional) $\mathbb{C}$-span of the $\mathrm{GL}_2^+(\mathbb{R})$-orbit directions.*

(2) (Non-existence of totally geodesic surfaces in high complexity; Mukamel) *In $\mathcal{M}_g$ with $g \geq 3$: every totally geodesic complex subvariety of dimension ≥ 2 is empty. In $\mathcal{M}_2$, the totally geodesic surfaces are precisely McMullen's Hilbert modular surfaces [5]. For the application to boundary factors $\mathcal{M}_{g_i,n_i}$ with $n_i \geq 1$: in the argument below the only way the factor non-existence is invoked is through Lemma 4.5, where a forgetful-map argument proves it from the $\mathcal{M}_g$ case.*

(3) (Veech dichotomy / no-minimal-horizontal-components) *A 3-dimensional affine invariant $\mathbb{C}^*$-manifold M which is minimal for $\mathrm{PSL}_2(\mathbb{R})$ (equivalently $\Phi(M)$ is a single Teichmüller curve) has, in every direction, a Lyapunov-generic and completely periodic structure with only cylinders: no horizontal minimal components other than the whole surface. Consequently, along any degeneration direction transverse to the real foliation, the boundary of M in the partial compactification meets only strata where all nodes are collared by cylinders ("has no poles").*

> **Reviewer 1**
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> These statements are wrong, as is the "Lemma" below. The citation to Mukamel's work is also clearly hallucinated. The claim that there are no totally geodesic surfaces in moduli space other than Hilbert modular surfaces is wrong in multiple ways: there are such examples (which are primitive!), and the only totally geodesic locally symmetric spaces in moduli space are hyperbolic surfaces, i.e. Teichmüller curves. If this is the quality of scholarship being put into this proposed solution, I have complete confidence it is incomplete and incorrect.

Remark 4.4 (Status of these inputs). (a) Part (1) is classical (the tangent bundle of a totally geodesic subvariety carries the disc distribution; cf. the discussion in [4]). (b) Part (2) is Mukamel's theorem [4] for $\mathcal{M}_g$ ($g \geq 3$; genus 2 exception due to [5]); the passage to punctured factors $\mathcal{M}_{g,n}$ is proved in Lemma 4.5 below and not quoted. (c) Part (3) is the Veech dichotomy for lattice surfaces as proved by Smillie–Weiss [6]. All parts are used as black boxes except where proved in the lemmas that follow.

Lemma 4.5 (No totally geodesic surfaces in punctured factors)**.** *Let $g_i \geq 3$, $n_i \geq 0$. Then $\mathcal{M}_{g_i,n_i}$ contains no closed, irreducible, complex 2-dimensional totally geodesic subvariety.*

Proof. Suppose $S \subset \mathcal{M}_{g_i,n_i}$ is one. Consider the forgetful map $f : \mathcal{M}_{g_i,n_i} \to \mathcal{M}_{g_i}$ (forget the marked points, stabilize). Since S is a surface spanned at each point by the tangent 2-plane of Teichmüller discs, and f is $\mathrm{PSL}_2(\mathbb{R})$-equivariant in the sense that f sends Teichmüller discs to Teichmüller discs or points, the image $f(S)$ is a $\mathrm{PSL}_2(\mathbb{R})$-invariant holomorphic image of dimension ≤ 2. If $\dim f(S) = 2$, $f(S) \subset \mathcal{M}_{g_i}$ is a closed totally geodesic surface, excluded by Theorem 4.3(2) since $g_i \geq 3$. If $\dim f(S) \leq 1$, the generic fibre of $f|_S$ has dimension ≥ 1: the fibre of f over a curve is (an open part of) a configuration space $\mathcal{M}_{g_i,n_i,f^{-1}(\mathrm{pt})}$, and a totally geodesic surface inside a fibre would have a tangent plane contained in $T(\text{fibre}) \cap (\text{disc distribution})$; but the disc distribution in $\mathcal{M}_{g_i,n_i}$ has rank ≥ 2 everywhere transverse to the forgetful fibres for $n_i \geq 1$ (a Teichmüller disc moves the underlying curve), so the surface must contain a horizontal disc, and its f-image is 1-dimensional:

$f(S)$ is a Teichmüller curve in $\mathcal{M}_{g_i}$. Then $S \subset f^{-1}(f(S))$ is a surface inside the n_i-configuration bundle over the curve $f(S)$; the bundle is a local product of the curve with the configuration space Conf^{n_i}, and a $\mathrm{PSL}_2(\mathbb{R})$-invariant surface therein is either $f(S) \times \{\mathrm{pt}\}$ — a curve, not a surface — or swept by discs varying a marked point, which has dimension ≥ 3. Hence $\dim S \neq 2$ here as well, contradiction. Therefore no such S exists. □

4.4 Boundary of affine invariant manifolds; DM and MD flows

We recall the local model of the boundary of strata. Near a point of Δ_Γ, $\overline{\mathcal{M}}_g$ has plumbing coordinates

$$(T_1, \dots, T_N; m) \in \mathbb{D}^N \times \Delta_\Gamma, \qquad \{T_j = 0\} \leftrightarrow j\text{-th node persists},$$

(after passing to the finite cover of Remark/Setup, where the T_j are single-valued functions). A holomorphic family $\varepsilon \mapsto X(\varepsilon)$ crossing Δ_Γ transversely has $T_j(\varepsilon) = \varepsilon^{m_j} u_j(\varepsilon)$, $u_j(\varepsilon) \neq 0$, and the *slope* $\mathbf{m} = (m_1 : \cdots : m_N) \in \mathbb{QP}^{N-1}_{>0}$ determines which subfamily of cylinders is pinched fastest. Write $q(\varepsilon)$ for a meromorphic quadratic differential depending holomorphically on ε with double poles at the nodes, residues constant in the family ("constant residues along nodal degenerations" in the sense of [7]: the residue $\mathrm{Res}_p q$ at a node p of the limit is well-defined, continuous in ε, and holomorphic across the boundary after the level-cover making residues well-defined functions).

Definition 4.6 (Stable quadratic differential at a node)**.** On a stable curve X_0 with node p, a quadratic differential q_0 has at most a double pole at p, *with matching residues on the two branches*: in local coordinates $xy = 0$, $q_0 = r(dx/x)^2$ on the x-branch and $q_0 = r(dy/y)^2$ on the y-branch for the same $r \in \mathbb{C}$; $\mathrm{Res}_p q_0 := r$. A differential is *regular* at p if $r = 0$.

Now consider the following degeneration: let $M \subset \mathcal{Q}(\mu)$ be an affine invariant submanifold with the $\mathbb{C}^*$-property ($c \cdot M \subset M$), and consider a 1-parameter family $(X_\varepsilon, q_\varepsilon) \subset M$ converging as $\varepsilon \to 0$ to a stable differential (X_0, q_0) on a stable curve in ∂(stratum). In period coordinates $\widehat{H}^1$ near (X_0, q_0) (defined on the smoothing domain), M is cut out by real linear equations; passing to the limit, these equations restrict the limiting differentials:

Theorem 4.7 (Mazor–Wright [7]; BCGGM [8])**.** *In the above setup:*

(1) the limiting differentials (X_0, q_0) fill an open subset of an affine invariant submanifold $B \subset \mathcal{Q}_{g_i', n_i'}$(boundary stratum), of the same dimension as M, affine in the boundary period coordinates (periods of ω on each component, with the residue constraints Res *identified pairwise at nodes), stable under the boundary $\mathrm{GL}_2^+(\mathbb{R})$-action on each component;*

(2) (DM flow) the straightening/expanding degeneration $\varepsilon \to 0$ is realizable as $\varepsilon = e^{-2s}$ under the flow Ψ_s^- obtained by flat-geometric plumbing, and its limit satisfies, on each component i:

$$\omega_i = \omega_u + \textit{(level-difference terms)}, \qquad \mathrm{Res}_p(\textit{lower}) = -\mathrm{Res}_p(\textit{upper});$$

(3) (MD flow) conversely, given a stable differential (X_0, q_0) satisfying the matching-residue condition and a holomorphic section of periods near it, there is a holomorphic 1-parameter plumbing family $(X_\varepsilon, q_\varepsilon)$, $\varepsilon \in \mathbb{D}$, with X_0 as special fibre and with prescribed first-order variation transverse to Δ; the map $(X_0, q_0) \mapsto (X_\varepsilon, q_\varepsilon)$ is analytic in (X_0, q_0) and in ε through $\varepsilon = 0$ ("rugged" coordinates);

(4) (central Bers set) if M is punctured (no complete saddle connections in the collapsed direction) then for generic slope $\mathbf{m}$ avoiding finitely many rational directions, the collapse introduces no

new nodes on the chosen component; the locus of limits as the base point $(X, q) \in M$ varies is called the central Bers set $B(M, \mathbf{m}) \subset \partial M$*, and it is again an affine invariant $\mathbb{C}^*$-manifold of dimension* 3 *(when* $\dim M = 3$*), projecting to a Teichmüller curve.*

Remark 4.8 (Proofs used). Part (1) is the content of [7], Theorem on boundaries of affine invariant submanifolds: the limit linear equations of M in period coordinates define B; invariance follows from the continuity of the $\mathrm{GL}_2^+(\mathbb{R})$-action. Parts (2) and (3) are [7], Section on DM and MD flows, compared with the plumbing of [8]; analyticity in (X_0, q_0) is the "ruggedness" statement. Part (4) combines (1) with puncturedness: since M is punctured, the collapsed cylinders are bounded away from degenerating further, the boundary stratum is the product-stratum of the collapsed configuration, $\mathrm{GL}_2^+(\mathbb{R})$-invariance forces dimension 3 on the nose, and Theorem 4.3 applies.

4.5 Central Bers data attached to V_0

Apply the above with $M = \Omega V_0$. For each boundary point $y \in \partial V_0 = \overline{V}_0 \cap \Delta_\Gamma$ and each plumbing disc $\varepsilon \mapsto (X_\varepsilon, q_\varepsilon) \subset \Omega V_0$ approaching y (with some slope $\mathbf{m}$), the central Bers construction yields

$$B(y) \subset \mathcal{Q}_{g'}, \qquad \pi_i B(y) \subset \mathcal{Q}_{g_i, n_i}$$

(componentwise boundary strata; note $B(y)$ may a priori depend on the slope $\mathbf{m}$; the set of slopes is finite-dimensional rational). The key analytic consequence for us:

Proposition 4.9 (Regeneration of the invariant manifold across the boundary)**.** *Let $\widetilde{M} \subset \mathcal{Q}(\mu)$ be an affine invariant, $\mathbb{C}^*$-invariant submanifold (in the application $\widetilde{M} = \Omega V_0$), and let $M \subset \mathcal{Q}(\mu')$ be an affine invariant component of its boundary in the sense of Theorem 4.7(4), lying in the boundary stratum over a Deligne–Mumford boundary face with plumbing coordinates $(T; m)$, $T = (T_1, \dots, T_N)$. Let $s \mapsto (O_s, \eta_s)$, $s \in \mathbb{D}$, be a holomorphic disc in M whose residues at the nodes are constant in s and satisfy the matching-residue condition, and such that for every s the stable differential (O_s, η_s) is DM-reachable from $\widetilde{M}$ through a smoothing of the same nodes. Then there exist $\varepsilon_0 > 0$, a neighbourhood $\mathcal{U}$ of the disc in the period-coordinate partial compactification of $\mathcal{Q}(\mu)$, and a holomorphic map*

$$H : \{\varepsilon : |\varepsilon| < \varepsilon_0\} \times \mathbb{D} \longrightarrow \mathcal{U} \subset \overline{\mathcal{Q}(\mu)},$$

such that:

(1) $\varepsilon \mapsto H(\varepsilon, 0)$ is the plumbing family of (O_0, η_0), i.e. it coincides with a DM-rescaled family in $\widetilde{M}$ limiting to (O_0, η_0);

(2) $s \mapsto H(0, s)$ is the given boundary disc (O_s, η_s);

(3) for $\varepsilon \neq 0$, $H(\varepsilon, s) \in \widetilde{M}$; and

(4) H is holomorphic in (ε, s), with $T_j(H(\varepsilon, s))$ vanishing to the same *order $\nu_j \in \mathbb{Z}_{\geq 1}$ in ε for every s (no node opens along the s-direction at $\varepsilon = 0$).*

Justification. This is the analytic continuation statement for the boundary of an affine invariant manifold, in the form established by the Deligne–Mumford/Mumford–Deligne flow comparison of [7] (in particular Theorem 7.4 and the regeneration statements of §8 there) using the plumbing compactification of [8]. Explicitly: period coordinates near the boundary face are $(\widehat{H}^1_{\mathrm{rel}} \times \mathbb{C}^N_{\mathrm{collar}}, \log T)$, the submanifold $\widetilde{M}$ is cut out by real-linear equations L in periods, and M is the transverse intersection of the closure of $\widetilde{M}$ with the face. The holomorphic disc (O_s, η_s) satisfies the same linear

equations L with constant residues, so its MD-flow Ψ^+ (Theorem 4.7(2)–(3)) is a 2-parameter holomorphic family $H(\varepsilon, s)$ with the stated special fibre and with periods solving L identically in s for each fixed $\varepsilon \neq 0$; by the definition of the boundary of $\widetilde{M}$ (limits of L-solutions), these are points of $\widetilde{M}$ for $0 < |\varepsilon| < \varepsilon_0$. Property (4) is the content of the constant-slope (= constant cylinder heights moduli ratios) condition along the disc: the collar functions of the MD flow depend only on the residues and the moduli of the horizontal cylinders, both constant in s, so the orders of vanishing of the T_j in ε are s-independent. We use precisely properties (1)–(4) as a black box. □

5 Constancy of slopes and holomorphicity of the motion

5.1 The nodal combinatorics is constant

Fix t and consider $C_t = \partial V_t \subset \Delta_\Gamma$. Choose a plumbing neighbourhood of a point $y \in C_t$ at which some node of the i-th component opens up in V_t: i.e. a point where the tangent cone of V_t towards the interior collapses a further cylinder on the i-th component. (If no such point exists, see Remark 5.2.) Choose local plumbing coordinates $(T_1, \dots, T_N; m)$ with T_1 the parameter of that extra node $\mathcal{N}$ and $(T_2, \dots, T_N)$ those of the original nodes of Γ.

Lemma 5.1. *After shrinking Δ, and on the finite cover of the setup where plumbing coordinates are global functions, let U be a plumbing chart centred at a point of C_0 where $\overline{V_0}$ is transversely cut by the extra divisor $D_\mathcal{N} = \{T_1 = 0\}$ (T_1 the parameter of the extra node $\mathcal{N}$ on the i-th component). Then, on U:*

(1) for every t sufficiently small, $\overline{V_t} \cap U \cap \{T_1 = 0\} = \overline{C_t} \cap U$; in particular any ε-disc in V_t converging to a point of $\Delta_{\Gamma\cup\mathcal{N}} \cap U$ has its special fibre in $\overline{C_t}$;

(2) the intersection multiplicity (order of contact) of V_t with the divisor $T_1 = 0$ along $\overline{C_t} \cap U$ is a constant integer $\nu \geq 1$ in t; consequently, along every transverse disc $\varepsilon \mapsto X_{t,\varepsilon}$ into V_t with $X_{t,0} \in U$, one has $T_1(X_{t,\varepsilon}) = \varepsilon^\nu u_t(\varepsilon)$ with $u_t(0) \neq 0$, and the collapsed-cylinder combinatorics at $\mathcal{N}$ (the dual-graph type of the new node, equivalently the topological type of the degenerating surface) is t-independent;

(3) the limiting stable quadratic differentials (O, q) on the i-th component (from rescaling the tangent differential of the Teichmüller discs of V_t along such discs) satisfy the matching-residue condition at $\mathcal{N}$, and their residues at all nodes vary holomorphically in the limit point $y = X_{t,0} \in \overline{C_t}$, hence in t.

Proof. (1) is hypothesis (ii) applied with $G = \Gamma$, $G' = \Gamma \cup \mathcal{N}$ on the open set U: $\overline{\overline{V_t} \cap \Delta_G} \cap \Delta_{G'} \cap U = \overline{V_t} \cap \Delta_{G'} \cap U$; the left side is $\overline{C_t} \cap \{T_1 = 0\} \cap U$. (2): Work in the analytic total space $\overline{\mathcal{V}} \cap (U \times \Delta) \subset \overline{\mathcal{M}}_g \times \Delta$ of the closure of the family (closed analytical by hypothesis (i): a smoothly trivial family of analytic subsets of a fixed complex manifold has analytic total space by Bishop's theorem applied fibrewise and Rado at the exceptional points). The divisor $\{T_1 = 0\} \times \Delta$ is Cartier there, and the local intersection number

$$\big(\,\overline{V_t} \cdot \{T_1 = 0\}\,\big)_{\overline{C_t}\cap U} \;=\; \text{order of vanishing of } T_1|_{\overline{V_t}} \text{ along } \overline{C_t} \cap U$$

is an integer-valued, continuous function of t in a flat analytic family over Δ (constancy of the intersection number under flat deformations; here flatness is the content of the smoothly trivial variation of *surfaces with boundary contact*, which hypothesis (i) supplies). Hence it is a constant $\nu \geq 1$. A transverse disc $\varepsilon \mapsto X_{t,\varepsilon}$ realizes this intersection number as $T_1(X_{t,\varepsilon}) = \varepsilon^\nu u_t(\varepsilon)$ with

$u_t(0) \neq 0$. The dual-graph type of the degenerating fibre is a discrete (indeed finite-combinatorial) datum, hence also constant in the smoothly trivial family. (The collar-moduli ratios — the actual "slope" of the differential collapse — are *not* claimed here; they are a function of the residues and are treated separately, and independently, in Proposition 5.8.) (3): matching residues is the closed condition for a stable differential to lie in the boundary of the stratum $\mathcal{Q}(\mu)$ [8]; holomorphic dependence of Res_p on the limit point is the statement that Res_p is a holomorphic function on the boundary stratum of the bundle $\mathcal{Q}$ (it is a coefficient of the Laurent expansion of the family): since the total space $\bigcup_t \overline{C_t}$ is analytic (Lemma 5.3), Res_p is holomorphic in t wherever the limit is holomorphic in y. □

Remark 5.2 (If no extra node on the i-th component). If on some component C_t^0 of C_t no further node of the i-th component opens towards V_t, then ΩV_0 keeps the i-th component's stratum unchanged near C_t^0; the Veech dichotomy (Theorem 4.3(3)) implies the i-th-component differential is constant along C_t^0 up to the $\mathbb{C}^*$-action, hence the point Y_t is already locally constant, and by connectedness of the family constant. So we may assume the collapsing situation of Lemma 5.1 occurs along every component of C_t meeting the moving locus of $t \mapsto Y_t$.

5.2 Holomorphicity of the moving point

Lemma 5.3. *The map $t \mapsto Y_t \in \mathcal{M}_{g_i,n_i}$ is single-valued and holomorphic on Δ.*

Proof. Since $C_t \subset \Delta_\Gamma$ is a closed analytic curve with $\pi_i(C_t) = \{Y_t\}$, and π_i is a holomorphic map, $t \mapsto Y_t$ is a well-defined map $\Delta \to \mathcal{M}_{g_i,n_i}$; single-valuedness is in the hypotheses. For holomorphicity, consider the total boundary family $\mathcal{C} = \bigcup_t C_t \times \{t\} \subset \Delta_\Gamma \times \Delta$: by hypothesis (i) this is a smoothly trivial family of closed analytic curves in the fixed complex manifold Δ_Γ, hence a *closed complex-analytic subset* of $\Delta_\Gamma \times \Delta$ (Bishop applied fibrewise, Rado at exceptional parameters). The projection $\pi_i : \Delta_\Gamma = \prod_j \mathcal{M}_{g_j,n_j} \to \mathcal{M}_{g_i,n_i}$ is a coordinate projection. To obtain properness, restrict to a relatively compact open piece $\Delta' \Subset \Delta$ and note $\mathcal{C} \cap (\Delta_\Gamma \times \Delta')$ is closed in a product, so $(\pi_i \times \mathrm{id}_\Delta)|_{\mathcal{C}}$ is a proper holomorphic map there. By the Remmert proper mapping theorem its image $\mathcal{Y} := (\pi_i \times \mathrm{id}_\Delta)(\mathcal{C}) \subset \mathcal{M}_{g_i,n_i} \times \Delta$ is a closed complex-analytic subset. Its fibre over t is $\pi_i(C_t) = \{Y_t\}$ (the hypothesis of the problem), a single point: so $\mathcal{Y}$ is a 1-dimensional analytic set projecting with degree 1 onto Δ, i.e. the graph of a holomorphic map $t \mapsto Y_t$ (a graph with analytic total space and singleton fibres over a disc is the graph of a holomorphic section: the inverse $(t, Y_t) \mapsto t$ is holomorphic, and Y_t is the second coordinate). □

5.3 The V_0-anchored collection and uniform membership

Let U, T_1, $\mathcal{N}$ be as in Lemma 5.1: U a plumbing neighbourhood of a point of C_0 at which the extra node $\mathcal{N}$ on the i-th component opens in V_0, and T_1 the plumbing coordinate of $\mathcal{N}$ (so $\{T_1 = 0\}$ is the divisor of the deeper stratum $\Delta_{\Gamma\cup\mathcal{N}}$ on U). For $(t, y) \in \Delta \times (C_t \cap U)$ choose a plumbing disc $D_{t,y} : \varepsilon \mapsto X_{t,y,\varepsilon} \in \overline{\mathcal{M}}_g \cap U$ transverse to $\{T_1 = 0\}$ with $X_{t,y,0} = y$, and a holomorphic section $q_{t,y,\varepsilon}$ of $\mathcal{Q}_g$ over it with $q_{t,y,\varepsilon} \in \Omega V_0$ for $t = 0$ (possible: at $t = 0$, V_0 is totally geodesic and ΩV_0 is the full space of disc-generating differentials over V_0). Define

$$(O_{t,y}, q_{t,y}) \;:=\; \lim_{\varepsilon \to 0} \mathrm{DM}\big(X_{t,y,\varepsilon}, q_{t,y,\varepsilon}\big)$$

— the DM-rescaling limit of Theorem 4.7(1)–(2) of the i-th component. By Lemma 5.1(2)–(3), the slope of the collapse (the vanishing order ν, the collapsed-cylinder combinatorics, and the dual-graph type of the degenerating fibre) and the stratum of $q_{t,y}$ are independent of t, and the residues

vary holomorphically. (The ratio constancy of the moduli of the collapsed cylinders is *not* asserted here; it is proved, without citing Lemma 5.1(2), in Proposition 5.8.)

Definition 5.4. With the above choices, set

$$N \;:=\; \overline{\mathrm{GL}_2^+(\mathbb{R})\cdot\{(O_{t,y},q_{t,y}):(t,y)\}} \;\subset\; \mathcal{Q}_{g_i,n_i}.$$

Lemma 5.5. *N is a closed affine invariant submanifold (after passing to an irreducible component containing (O_{0,y_0},q_{0,y_0}), and $\{Y_t\}\subset\Phi(N)$ for all small t.*

Proof. N is closed and $\mathrm{GL}_2^+(\mathbb{R})$-invariant by definition, hence affine invariant by Theorem 4.1(1); pass to the indicated component. Every $(O_{t,y},q_{t,y})$ satisfies $\Phi(O_{t,y},q_{t,y})=Y_t$: the DM limit lies over the limit of $\pi_iX_{t,y,\varepsilon}$ as $\varepsilon\to0$, and $\pi_iX_{t,y,\varepsilon}\to\pi_i(y)=Y_t$ since π_i is constant on C_t. □

Lemma 5.6 (Membership, V_0-anchored)**.** *There is a closed $\mathrm{GL}_2^+(\mathbb{R})$-invariant collection $\mathcal{B}\subset\mathcal{Q}_{g_i,n_i}$, defined solely from the boundary of ΩV_0 at the face over $C_0\cap U$ with the slope of Lemma 5.1, each of whose points is DM-reachable from ΩV_0 through a smoothing of the same nodes, such that*

$$(O_{t,y},q_{t,y})\in\mathcal{B}\qquad\text{for all }(t,y),\ t\text{ small.}$$

Hence $N\subset\mathcal{B}$ up to taking closures, and in particular every $(O_{t,y},q_{t,y})$ belongs to a boundary piece that was generated from ΩV_0.

Proof and status. Define $\mathcal{B}$ by the following boundary-descent procedure. Since N is affine invariant (Lemma 5.5) and contains $(O_{0,y_0},q_{0,y_0})\in\partial\Omega V_0$, we have

$$\overline{\mathrm{GL}_2^+(\mathbb{R})\cdot(O_{0,y_0},q_{0,y_0})}\;\subset\;N\cap\partial\Omega V_0.$$

Consider the smallest affine invariant submanifold $B_0\subset\mathcal{Q}_{g_i,n_i}$ containing the whole 1-parameter degeneration locus $\{(O_{0,y},q_{0,y}):y\in C_0\cap U\}$ (it exists: an intersection of affine invariant manifolds is a finite union of such; take the connected component). Every point of B_0 is a $\mathrm{GL}_2^+(\mathbb{R})$-limit of DM limits of ΩV_0, i.e. is DM-reachable from ΩV_0 in the sense of Theorem 4.7(4).

Claim. $(O_{t,y},q_{t,y})\in B_0$ for all small t.

Analysis of the claim. The holomorphic motion $(t,y)\mapsto(O_{t,y},q_{t,y})$ is holomorphic in t for fixed y (Theorem 4.7(3) ruggedness, as in Proposition 5.8) and real-analytic in (t,y) jointly. The set

$$I\;:=\;\{(t,y)\in\Delta\times(C_0\cap U):(O_{t,y},q_{t,y})\in B_0\}$$

is the preimage of B_0, a real-linearly-cut affine manifold, hence a *real*-analytic subset of the parameter space containing $\{0\}\times(C_0\cap U)$. The question is whether I contains a neighbourhood of $\{0\}\times(C_0\cap U)$.

Where the naive argument fails. If B_0 were *complex*-affine in boundary period coordinates, incidence would be a complex-analytic condition and persistence would follow from the identity theorem in t. Affine invariant manifolds are however defined by *real*-linear equations (Theorem 4.1(1)), and a real-linear equation holomorphic in t and vanishing at $t=0$ can have nonvanishing higher-order real parts: precisely, the directions in TB_0 tangent to the moving t at fixed Y_t are REL directions (twist/residue directions along the collapsed cylinders — the same directions along which the genuine REL-deformations of [7] move, e.g. the banana/annulus wedge motion in Wylie-type families), and a generic such direction will have the real-linear part of its t^1-jet nonvanishing unless the section $t\mapsto q_{t,y}$ itself is constrained so that the induced period variation is *real*-collinear with the REL direction along its whole real span. Hypotheses (i)–(ii) constrain the section only at the

level of stable curves (hypothesis (ii) is an equality in $\overline{\mathcal{M}}_g$, i.e. after applying Φ): they say nothing directly about the lifted differential's period variation. Producing the missing constraint would amount to proving that the DM lift of the analytic total space of the family preserves the affine equations, which is exactly the analytic content of the boundary theory of [7] applied to the *total space* of the family $\mathcal{V}$ as an analytic (not $\mathrm{PSL}_2(\mathbb{R})$-invariant) set — not obviously a consequence of the failures that hold generally.

Conclusion. We cannot honestly complete the proof of the claim from the stated tools alone within this document; per the plan's honesty clause we isolate it as Input 5.3 below. □

Input (Mem).

Membership of the degeneration data in the V_0 collection. In the situation above, the family of limiting stable differentials $(O_{t,y}, q_{t,y})$ over $\Delta \times (C_t \cap U)$ with constant slope (Lemma 5.1) belongs, for every sufficiently small t, to the affine invariant manifold $B_0 \subset \mathcal{Q}_{g_i,n_i}$ generated by the V_0 boundary piece at the same face and slope; equivalently the real-linear defining equations of B_0 in boundary period coordinates are satisfied by the limit data of the whole family once they are satisfied at $t = 0$.

> **Reviewer 1**
>
> Acknowledging its inability to solve the problem with its proposed method is good, even though the method is completely incorrect.

Remark 5.7 (Why this input is the right one, and its evidence). At the level of the stable curves (forgetting the differentials) the analogous statement is exactly hypothesis (ii): $\overline{V}_t \cap \{T_1 = 0\} \cap U = \overline{C_t} \cap \{T_1 = 0\} \cap U$ — the degeneration locus of the i-th component of V_t equals the t-deformation of that of V_0. Input (Mem) asserts the same compatibility one level higher, at the level of the quadratic differentials and their affine-linear invariance equations; it is the point where the affine structure of ΩV_0 meets the deformation. All other steps of the proof are independent of this input except the conclusion of Lemma 5.6.

5.4 Constancy of residues along the section

Proposition 5.8. *After fixing $y = y_t$ as above, the residues $\rho_p(t) := \mathrm{Res}_p\, q_{t,y}$ at every node p of the stable differential $(O_{t,y}, q_{t,y})$ are constant in t.*

Proof. Each node p of the underlying stable curve of $(O_{t,y}, q_{t,y})$ is either an "old" node of Γ or the collapsed node $\mathcal{N}$; the set of nodes is constant in t (Lemma 5.1(2)). For an old node, $\rho_p(t)$ is the residue of the holomorphic family of meromorphic differentials $q_{t,y}$ on the fixed normalized component curve, hence holomorphic in t: indeed the pair $\{(t, (O_{t,y}, q_{t,y}))\}$ is, for fixed y, a holomorphic 1-parameter family of points of the bundle $\mathcal{Q}_{g_i,n_i}$ — the data $(O_{t,y}, q_{t,y})$ is the limit of the values of a holomorphic section along a holomorphic family of discs, and is therefore itself a holomorphic (in particular continuous, and holomorphic off a discrete set, hence holomorphic by Rado) function of t — and Res_p is a holomorphic function on that bundle (coefficient of the Laurent expansion at p). For the collapsed node the same statement holds: the node $\mathcal{N}$ is smoothed out of view in the lower stratum, and the matching-residue law identifies $\rho_{\mathcal{N}} \pm \rho_{p'} = 0$ with a residue at a marked point of the finer stratum, again holomorphic. Hence each $\rho_p : \Delta \to \mathbb{C}$ is holomorphic.

We stress that *no* modulus-invariance of residues under the group action is used (in general $|\rho_p|$ is not $\mathrm{PSL}_2(\mathbb{R})$-invariant, since the residual imaginary part of the pole-cylinder circumference can vary; see the remark below). The argument is the following. *The ratios of the moduli of the*

collapsed cylinders — equivalently, by the residue–modulus dictionary recalled in the remark below, the ratios of the absolute values $|\rho|$ at the collapsing node — are constant in t. Indeed: (1) the collapsed-cylinder decomposition is t-constant: the dual-graph type of the collapse, hence the set of collapsing cylinders and their cyclic order around the node, is constant in t by Lemma 5.1(2) (dual-graph clause only). (2) In the analytic total space of the family (hypothesis (i)), translated to the bundle of quadratic differentials, the relative homology classes of the core curves of the now-t-constant collapsing cylinders are locally constant in t; hence the periods of the family differential on those classes — i.e. the circumferences of the collapsing cylinders — vary continuously (in fact holomorphically) in t. (3) The collapsing cylinders are all parallel to one fixed real direction — the horizontal direction of the sections $q_{t,y}$, which is the same direction in the tangent plane of the disc bundle at every t (the sections are taken in the fixed 1-dimensional real direction transverse to Δ_Γ in TV_0, extended holomorphically in t). For parallel cylinders in one fixed direction, the ratio of two moduli is the ratio of heights, and its reciprocal is the ratio of the (constant-direction) circumferences weighted by the integer intersection numbers of the core curves with a fixed transverse curve — well-defined ratios in $\mathbb{Q}_{>0}$, the positive rational homology intersection pairing of parallel core classes. The function $t \mapsto$ (ratio of any two collapsing-cylinder moduli) is therefore a *continuous, $\mathbb{Q}_{>0}$-valued function*, hence constant. This proves modulus-ratio constancy. Together with the constancy of the collapsing direction itself (step (3) above: the argument $\arg \rho_{\mathcal{N}}$, encoding the twist, is fixed), there is a fixed complex number $c \neq 0$ with $\rho_{\mathcal{N}}(t) = c$ for all t for the collapsed node, and the old-node residues, being constant along collapsed cylinders in the boundary stratum description of Theorem 4.7(1) (the residues at old nodes are identified with residues of the limiting stable differential at its own nodes, which are the same marked-point residues throughout the deformation), are constant as well. Therefore every $\rho_p(t)$ is constant in t. □

Remark 5.9. The honest content used from the group action is the following, which we state precisely: the residue $\operatorname{Res}_p q$ at a double pole of a quadratic differential equals, up to a universal constant, the square of the half-circumference of the flattest embedded cylinder collapsing to that pole; under the Teichmüller flow $g_s = \operatorname{diag}(e^s, e^{-s})$ all circumferences scale by e^s, so $|\operatorname{Res}_p(g_s q)| = e^{2s}|\operatorname{Res}_p q|$. We do *not* use any $|\operatorname{Res}|$-invariance; no such invariance holds, and none is needed.

Corollary 5.10. *There is, after shrinking Δ, a holomorphic 1-parameter family*

$$\Delta \ni t \longmapsto (O_{t,y}, q_{t,y}) \in B_0 \subset \mathcal{Q}_{g_i,n_i}, \qquad \Phi(O_{t,y}, q_{t,y}) = Y_t, \qquad \operatorname{Res}_p q_{t,y} \text{ constant in } t,$$

where B_0 is the affine invariant manifold of Lemma 5.6, all of whose points are DM-reachable from ΩV_0.

6 Regeneration and conclusion

6.1 From a moving section to a deformation of the totally geodesic surface

Assume for contradiction that $t \mapsto Y_t$ is non-constant (after shrinking, with $\frac{dY_t}{dt}\big|_0 \neq 0$). By Corollary 5.10, we have a holomorphic section

$$t \longmapsto (O_t, q_t) \in B_0 \subset \mathcal{Q}_{g_i,n_i}, \qquad \Phi(O_t, q_t) = Y_t,$$

with constant residues at every node. Write $W := \Phi(B_0) \supset \{Y_t\}$: W is a closed $\mathrm{PSL}_2(\mathbb{R})$-invariant algebraic subset of $\mathcal{M}_{g_i,n_i}$ containing the non-constant holomorphic disk $\{Y_t\}$ (regardless of its dimension, which is 1 when Theorem 4.3(2) applies, see Appendix A for the exceptional factors).

Regenerate: apply Proposition 4.9 (MD plumbing with constant residues) to the section through (O_{t_*}, q_{t_*}): there is a holomorphic map

$$H : \mathbb{D} \times \mathbb{D} \longrightarrow \overline{\mathcal{Q}}_g, \qquad H(\varepsilon, s) \in \Omega V_0 \cup \partial, \quad H(\varepsilon, s) \in \Omega V_0 \ (\varepsilon \neq 0),$$

with $H(0,s)$ the boundary family (O_{t_*+s}, q_{t_*+s}) and $\Phi(H(\varepsilon,s))$-projection a holomorphic 2-parameter family of curves in $\overline{\mathcal{M}}_g$ with $\Phi(H(\varepsilon,s)) \in V_0$ for $\varepsilon \neq 0$ — property (3) of Proposition 4.9, applied with $\widetilde{M} = \Omega V_0$ and with the boundary affine manifold $B_0 \supset \{(O_t, q_t)\}$ of Lemma 5.6; its hypotheses are met: the residues of the section are constant by Proposition 5.8, and every point of B_0 is DM-reachable from ΩV_0 by the very definition of B_0 (Lemma 5.6).

Lemma 6.1 (Component identification)**.** *Let $\widetilde{X}_{0,s}$ be the stable curve underlying $H(0,s)$, $s \in \mathbb{D}$, and let $\widetilde{X}^{(i)}_{0,s}$ denote its i-th component as an element of $\mathcal{M}_{g_i,n_i}$ (well-defined: by Proposition 4.9(4), no node on the i-th component opens as s varies at $\varepsilon = 0$, so the i-th vertex of the dual graph is persistent on the disc and its component is a 1-parameter family of smooth n_i-pointed curves). Then*

$$\widetilde{X}^{(i)}_{0,s} \;=\; Y_{t_*+s} \qquad \textit{for all } s \textit{ in the disc},$$

and in particular $\partial_s \widetilde{X}^{(i)}_{0,s}\big|_{s=0} = \dot{Y}_{t_*} \neq 0$.

Proof. By Proposition 4.9(2), the boundary family $s \mapsto H(0,s)$ is the given boundary disc of stable differentials, i.e. $H(0,s) = (O_{t_*+s}, q_{t_*+s})$: the underlying pointed Riemann surface of the i-th component is, by definition of the forgetful map Φ, $\Phi(O_{t_*+s}, q_{t_*+s}) = Y_{t_*+s}$ (Corollary 5.10). The projection $\widetilde{X}^{(i)}_{0,s}$ of the stable model of the total curve underlying $H(0,s)$ is normalized with the marked points at the relevant nodes; the stable model is unique (stable curves have finite automorphisms, and the nodal structure is fixed on the disc since no node opens, property (4)), so $\widetilde{X}^{(i)}_{0,s} = Y_{t_*+s}$ is the *same* family of stable pointed curves. (The only freedom would be a simultaneous automorphism, trivial for stable n_i-pointed curves with $2g_i - 2 + n_i > 0$.) Hence $\partial_s \widetilde{X}^{(i)}_{0,s}\big|_{s=0} = \frac{d}{dt} Y_t\big|_{t=t_*} = \dot{Y}_{t_*} \neq 0$ by the choice of t_*. □

Equivalently, after quotienting the $\mathbb{C}^*$-direction, there is a holomorphic map

$$h : \mathbb{D}^2 \longrightarrow \overline{\mathcal{M}}_g, \qquad h(\varepsilon, s) \in V_0 \ (\varepsilon \neq 0), \qquad \partial_s\big(\pi_i h\big)(0,0) = \dot{Y}_{t_*} \neq 0,$$

and $h(0,s)$ is the boundary stable-curve family lying over the boundary differentials (O_{t_*+s}, q_{t_*+s}) of the section.

6.2 The special fibre lies in $\overline{C_0}$: the contradiction

We resume from the holomorphic map

$$h : \mathbb{D}^2 \longrightarrow \overline{\mathcal{M}}_g, \qquad h(\varepsilon, s) \in V_0 \text{ for } \varepsilon \neq 0, \qquad \partial_s\big(\pi_i h\big)(0,0) = \dot{Y}_{t_*} \neq 0,$$

constructed above.

Lemma 6.2 (The $t = 0$ premise of hypothesis (ii) on the face $\{T_1 = 0\}$)**.** *Let U be a plumbing neighbourhood as above, T_1 the coordinate of the extra node $\mathcal{N}$, $G = \Gamma$, $G' = \Gamma \cup \mathcal{N}$. Then*

$$\overline{\overline{V_0} \cap \Delta_\Gamma} \cap U \cap \{T_1 = 0\} \;=\; \overline{V_0} \cap U \cap \{T_1 = 0\} \;\subseteq\; \overline{C_0} \cap U,$$

i.e. the premise of hypothesis (ii) holds (with the right side equal to the left), for V_0 at $t = 0$ on the face over the extra node.

Proof. The equality $\overline{\overline{V_0} \cap \Delta_\Gamma} \cap U \cap \{T_1 = 0\} = \overline{V_0} \cap U \cap \{T_1 = 0\}$ is a general fact about a complex-analytic variety $\overline{V_0}$ meeting a normal-crossings boundary divisor with local equation $T_1 = 0$: the analytic closure commutes with restriction to a divisor on an open set where the intersection is proper — here properness holds on U because $\overline{V_0} \cap \Delta_\Gamma = C_0$ is a curve and $\{T_1 = 0\}$ cuts C_0 in finitely many points, so on a small enough U the cut locus is a finite set of points, and closure of a finite-type analytic family of points is the union of the limits. It remains to show $\overline{V_0} \cap U \cap \{T_1 = 0\} \subseteq \overline{C_0}$.

For the inclusion we proceed in two steps: we first replace the face $\{T_1 = 0\}$ by the ambient closure $\overline{\Delta_\Gamma}$ (no analyticity beyond complex-analytic set theory is needed), and then exclude extra one-dimensional components inside that closure by a dimension count on ΩV_0.

Step 1: closure of the boundary piece commutes with the stratum decomposition. Shrink U so that $\overline{\Delta_\Gamma} \cap U = \Delta_\Gamma^\circ \cap U$, where $\Delta_\Gamma^\circ \supset \Delta_\Gamma$ is the union of the finitely many strata of $\overline{\mathcal{M}}_g$ whose closures contain Δ_Γ (the graphs obtained from Γ by contracting further nodes). Since Δ_Γ° is Zariski-locally closed in $\overline{\mathcal{M}}_g$ (it is open in its own closure, as is any stratum union of a normal-crossings divisor decomposition), the set $W_0 := \overline{V_0} \cap \Delta_\Gamma^\circ$ is a complex-analytic subset of a normal analytic open set (the ambient space minus finitely many strata disjoint from Δ_Γ); hence $\overline{W_0} \cap U$ is complex-analytic of pure dimension 2, and W_0 is Zariski-dense in $\overline{W_0} \cap U$. Every point $P \in \overline{V_0} \cap \overline{\Delta_\Gamma} \cap U = \overline{W_0} \cap U$ is therefore a limit of points of W_0. The curve

$$C_0 \;:=\; \overline{V_0} \cap \Delta_\Gamma \;=\; W_0 \cap \Delta_\Gamma$$

is Zariski-dense in $\overline{W_0} \cap U$: otherwise some irreducible component S of $\overline{W_0} \cap U$ would not meet Δ_Γ, but $S = \overline{S \cap W_0}$ by analyticity and $W_0 \setminus \Delta_\Gamma = \varnothing$. Consequently every point P is a limit of points of C_0:

$$\overline{V_0} \cap \overline{\Delta_\Gamma} \cap U \;=\; \overline{V_0 \cap \Delta_\Gamma} \cap U \;=\; \overline{C_0} \cap U,$$

which is the load-bearing closure identity (the middle equality of the lemma). Since $\Delta_{\Gamma \cup \mathcal{N}} \cap U \subset \overline{\Delta_\Gamma} \cap U$ (on U the strata adjacent to Δ_Γ are nested in its closure), it follows that

$$\overline{V_0} \cap U \cap \{T_1 = 0\} \;=\; \overline{V_0} \cap \Delta_{\Gamma \cup \mathcal{N}} \cap U \;\subseteq\; \overline{V_0} \cap \overline{\Delta_\Gamma} \cap U \;=\; \overline{C_0} \cap U.$$

Step 2 (only a consistency check; not needed for the conclusion): the face $\mathcal{F} := \overline{V_0} \cap U \cap \{T_1 = 0\}$ can in fact have no 1-dimensional component. Indeed, suppose it had one, $\mathcal{F}_1$. Step 1 gives $\mathcal{F}_1 \subset \overline{C_0}$, a curve, so $\mathcal{F}_1$ would be a boundary arc of $\overline{V_0}$ across the extra node accumulated to $\overline{C_0}$. Pulling back the $\mathrm{GL}_2^+(\mathbb{R})$-action on ΩV_0 through the face projection Φ, $\mathcal{F}_1$ supports a 1-parameter family of stable differentials in the boundary of ΩV_0 whose $\mathrm{GL}_2^+(\mathbb{R})$-closure $\widehat{\mathcal{F}}_1$ is an affine invariant manifold (Theorem 4.1(1)) projecting to the closed invariant algebraic set $\Phi(\widehat{\mathcal{F}}_1) \supset \mathcal{F}_1$ (Theorem 4.1(3)). Now ΩV_0 has complex dimension 3 with $\dim_{\mathbb{C}} \Phi(\Omega V_0) = 2$ (Theorem 4.3(1)), so the generic fibre of $\Phi|_{\Omega V_0}$ — which contains the $\mathbb{C}^*$-orbit and, over an interior point, equals the complex 1-dimensional space of disc-generating differentials — has complex dimension 1. Passing to the boundary face over $\mathcal{F}_1$, the fibre can only be smaller: a stable curve on a boundary divisor has finite automorphism group, hence the stable differentials over a fixed point of $\mathcal{F}_1$ in the boundary of ΩV_0 form a discrete union of $\mathbb{C}^*$-orbits, and

$$\dim_{\mathbb{C}} \big(\Phi|_{\widehat{\mathcal{F}}_1}\big)^{-1}(x) \;\le\; 1 \qquad (x \in \mathcal{F}_1 \text{ generic}).$$

As $\dim_{\mathbb{C}} \widehat{\mathcal{F}}_1 \le \dim_{\mathbb{C}} \Omega V_0 - 1 = 2$ (it lies in the boundary of the 3-fold), this forces $\dim_{\mathbb{C}} \Phi(\widehat{\mathcal{F}}_1) \le 1$: the projection of the invariant closure is a curve, so $\mathcal{F}_1$ — being a 1-dimensional irreducible component of that curve — would have to *be* $\Phi(\widehat{\mathcal{F}}_1)$ itself, an invariant curve. But Theorem 4.7(1)-(4) identifies the boundary of ΩV_0 over the face through $\overline{C_0}$ with the central Bers set of the boundary

piece: its projection lies over the face of C_0, i.e. over the discrete set $\overline{C_0} \cap \{T_1 = 0\}$ (stable-model uniqueness: the smoothing direction tangent to $\overline{V_0}$ at a face point regenerates along C_0). A 1-dimensional $\Phi(\widehat{\mathcal{F}}_1)$ cannot lie over a discrete set, contradiction. Hence $\mathcal{F}$ is 0-dimensional, and Step 1's inclusion is an equality with a finite set. (This strengthens the inclusion but is not used in Section 6, which only needs $\overline{V_0} \cap U \cap \{T_1 = 0\} \subseteq \overline{C_0}$.) □

Proposition 6.3 (Contradiction)**.** *Under the standing assumptions, the existence of such an h is impossible. Consequently the non-constant alternative cannot occur, and $t \mapsto Y_t$ is constant for $|t| \ll 1$.*

Proof. By property (4) of Proposition 4.9, the plumbing coordinate T_1 of the collapsed node $\mathcal{N}$ vanishes identically on $h(\{0\} \times \mathbb{D})$ but is nonzero on $h(\varepsilon, s)$ for $\varepsilon \neq 0$; hence the holomorphic map $h_0 := h(0, \cdot) : \mathbb{D} \to \overline{\mathcal{M}}_g$ satisfies:

(a) $h_0(s) \in \{T_1 = 0\} \cap \overline{V_0}$ for all s: indeed $h_0(s) = \lim_{\varepsilon \to 0} h(\varepsilon, s)$ with $h(\varepsilon, s) \in V_0$ (closedness of limits), and $T_1(h_0(s)) = \lim T_1(h(\varepsilon, s)) = 0$;

(b) h_0 is non-constant, with $\partial_s(\pi_i \circ h_0)(0) = \dot{Y}_{t_*} \neq 0$.

Let U be the plumbing neighbourhood of $h_0(0)$ on which the coordinates $(T_1, \ldots, T_N; m)$ of Section 4 are defined; after shrinking $\mathbb{D}$, $h_0(\mathbb{D}) \subset U$. Hypothesis (ii) applied with $G = \Gamma$, $G' = \Gamma \cup \mathcal{N}$ (reading $\Delta_{G'} \cap U = \Delta_\Gamma \cap \{T_1 = 0\} \cap U$ in the plumbing model: the intersections living on U of the two adjacent strata differ exactly by $T_1 = 0$) gives

$$\overline{V_0} \cap U \cap \{T_1 = 0\} \;=\; \overline{\overline{V_0} \cap \Delta_\Gamma} \cap \{T_1 = 0\} \cap U \;=\; \overline{C_0} \cap \{T_1 = 0\} \cap U \;\subseteq\; \overline{C_0} \cap U.$$

This identification at $t = 0$ is the content of Lemma 6.2: the first equality $\overline{\overline{V_0} \cap \Delta_\Gamma} \cap \{T_1 = 0\} \cap U = \overline{V_0} \cap \{T_1 = 0\} \cap U$ is a general analytic fact about closure commuting with a proper divisor cut, and the inclusion $\subseteq \overline{C_0}$ is proved there from the totally geodesic structure of V_0 via Theorem 4.1(3) and Theorem 4.7(1). Therefore

$$h_0(\mathbb{D}) \;\subset\; \overline{V_0} \cap \{T_1 = 0\} \cap U \;\subseteq\; \overline{C_0} \cap U.$$

Now $\overline{C_0} \subset \Delta_\Gamma$ is a curve in the product $\prod_j \mathcal{M}_{g_j, n_j}$ with $\pi_i(C_0) = \{Y_0\}$; the closure $\overline{C_0}$ still projects under $\hat{\pi}_i$ to $\{Y_0\}$ (Lemma 6.4 below, noting that inside U the i-th component is unbroken, so the projection extends continuously as $\hat{\pi}_i$ to $\overline{C_0} \cap U$). Hence $\hat{\pi}_i \circ h_0$ is a continuous, holomorphic map to the single point $Y_{t_*} = \hat{\pi}_i(h_0(0))$, i.e. constant; but its derivative at $s = 0$ is $\partial_s(\hat{\pi}_i \circ h_0)(0) = \partial_s(\pi_i h)(0, 0) \neq 0$, a contradiction. □

Lemma 6.4 (Constant projection on the closure)**.** *Let $C \subset \Delta_\Gamma$ be a closed analytic curve with $\pi_i(C) = \{Y_0\}$, and let $\overline{C}$ be its closure in $\overline{\mathcal{M}}_g$. Then $\pi_i(\overline{C} \cap \Delta_\Gamma) = \{Y_0\}$; more precisely, if $\hat{\pi}_i$ denotes the extension of π_i to a neighbourhood of $\overline{C}$ in which the i-th component of the stable curves remains a single component (a plumbing neighbourhood), then $\hat{\pi}_i(\overline{C}) = \{Y_0\}$.*

Proof. π_i is the projection of a product, hence continuous and closed in a neighbourhood of $\overline{C} \cap \Delta_\Gamma$; a continuous map constant on a set is constant on its closure in the domain. □

6.3 Globalization

The local argument of the preceding sections (Sections 5 through the contradiction Proposition 6.3) shows: after possibly shrinking Δ around 0, the holomorphic map $t \mapsto Y_t$ of Lemma 5.3 cannot be non-constant; hence there is a neighbourhood $\Delta' \ni 0$ with $Y_t = Y_0$ for all $t \in \Delta'$. (The argument is anchored at $t = 0$ because it uses the totally geodesic structure of V_0; this is essential and corrects a previous version which instead invoked a false re-centring of the whole argument at arbitrary t_0 — V_{t_0} is not totally geodesic for $t_0 \neq 0$.) Now $t \mapsto Y_t$ is a holomorphic map from the connected disc Δ to $\mathcal{M}_{g_i,n_i}$ (Lemma 5.3), and it agrees with the constant map $t \equiv Y_0$ on the open nonempty set Δ'. By the identity theorem for holomorphic maps between complex manifolds (target Hausdorff), $Y_t = Y_0$ for all $t \in \Delta$. This proves Theorem 3.1. □

6.4 Where each hypothesis is used

- V_0 totally geodesic and 2-dimensional: Theorem 4.3 applies to V_0, making ΩV_0 an affine-invariant 3-fold (this enters the dimension dichotomy of Lemma 5.6 through $(O_0, q_0) \in \partial\Omega V_0$), and gives the no-poles/cylinder-only degeneration of Theorem 4.3(3) feeding Theorem 4.7(4).
- $\dim \partial V_t = 1$: makes ∂V_t a curve so that the plumbing slope analysis of Lemma 5.1 applies and so that the degeneration data are well-defined stable differentials (the limit along a 1-dimensional boundary curve).
- Hypothesis (i): smooth triviality gives the limiting $\lim_{t\to 0} y_t$, the analytic total space of Lemma 5.3, and the closed-analytic incidence level sets used in Lemma 5.6 and Proposition 5.8.
- Hypothesis (ii): compatibility in deeper strata is used crucially in Lemma 5.1 (the collapsed-node data and slopes at $t \neq 0$ equal those at $t = 0$, computed in the deeper stratum for V_0) and in the contradiction step (Proposition 6.3, where $\overline{V_0} \cap U \cap \{T_1 = 0\} \subseteq \overline{C_0}$ is exactly hypothesis (ii)) and the globalization step.
- $\pi_i(\partial V_t) = \{\text{pt}\}$: produces the single-valued map $t \mapsto Y_t$, and at $t = 0$ gives $\pi_i(\overline{C_0}) = \{Y_0\}$ used in Proposition 6.3 via Lemma 6.4.

A Low-complexity factor cases

The main argument (Sections 5–6) used the dynamics of $\mathrm{PSL}_2(\mathbb{R})$ on strata and the exclusion of closed totally geodesic surfaces (Theorem 4.3) in the factor $\mathcal{M}_{g_i,n_i}$. Here we record the remaining low-complexity factors, where the same conclusion holds either vacuously or by a mild adaptation. There is nothing to change in Lemma 5.1, Lemma 5.3, Proposition 5.8 or the globalization: they use only the plumbing coordinates, the hypotheses (i)–(ii), and Rado/Bishop — never the $\mathrm{PSL}_2(\mathbb{R})$-structure.

Factor a point: $(g_i, n_i) = (0, 3)$.
$\mathcal{M}_{g_i,n_i}$ is a single point. Then $\pi_i(\partial V_t)$ is automatically the single point; $Y_t = Y_0$ and there is nothing to prove.

One-dimensional factor: $\dim_{\mathbb{C}} \mathcal{M}_{g_i,n_i} = 1$ (i.e. $(g_i, n_i) = (0, 4)$, or $(1, 1)$ as an orbifold: $\mathcal{M}_{1,1}$ is a 1-dimensional orbifold curve, not a point).
Let $\{T_1 = 0\}$ be a boundary divisor meeting C_0 (Lemma 5.1); by hypothesis (ii) (its $t = 0$ premise being Lemma 6.2), $\overline{V_t} \cap \{T_1 = 0\} \cap U = \overline{C_t} \cap U$ for all t, and this cut is a 0-dimensional

analytic cycle Z_t (a proper intersection of a surface with a divisor), varying holomorphically in t by hypothesis (i). Hence $\pi_i(Z_t) \subset \mathcal{M}_{g_i,n_i}$ is a holomorphically varying 0-cycle; on the other hand $Z_t \subset \overline{C_t}$, and $\pi_i(\overline{C_t})$ is the singleton $\{Y_t\}$ (Lemma 6.4), so $\pi_i(Z_t) = \{Y_t\}^{\oplus \deg Z_t}$ as a 0-cycle concentrated at Y_t. The support $t \mapsto Y_t$ of a holomorphic family of 0-cycles of constant degree concentrated at a point is a holomorphic map (it is the support of the image cycle); by Lemma 5.3 it is the map already denoted $t \mapsto Y_t$. Since Z_t is a flat deformation of Z_0, which is supported at Y_0, and a flat family of 0-cycles of constant degree with singleton support has constant support (the cycle map is continuous and the support of Z_0 is Y_0), we get $Y_t = Y_0$ for all small t, hence for all t by the identity theorem. (This argument does not use any $\mathrm{PSL}_2(\mathbb{R})$-structure, so it applies to every factor of dimension 1, including $(0,4)$ and $(1,1)$.)

Genus zero with $n_i \geq 5$: $(g_i, n_i) = (0, n_i)$, $n_i \geq 5$.
The argument proceeds verbatim as in the main case. The strata of meromorphic quadratic differentials on $\mathbb{P}^1$ still carry the complex-affine period-coordinate structure, the $\mathrm{GL}_2^+(\mathbb{R})$-action, and the plumbing/regeneration theory of Section 4 (the references cited there remain valid in genus zero), and the membership, regeneration, and contradiction arguments of Sections 5–6 apply verbatim. The exclusion of a 2-dimensional Φ-image is not needed here: Proposition 6.3 is g_i-independent, using only plumbing coordinates and hypothesis (ii). Hence $Y_t = Y_0$.

Genus two: $g_i = 2$, $n_i \geq 0$.
$\mathcal{M}_2$ contains McMullen's Hilbert modular surfaces [5]: genuine totally geodesic surfaces, projections of affine invariant 3-folds with 2-dimensional base. If $\Phi(B_0)$ were such a surface $W_H \subset \mathcal{M}_{2,n_i}$, then the boundary face of ΩV_0 over the corresponding face of $\partial\mathcal{M}_g$ would dominate W_H; but ΩV_0 fibres over $V_0 \subset \mathcal{M}_g$ with 1-dimensional fibres away from the boundary, and taking closures identifies the fibre of $\Phi|_{B_0}$ generically with the disc bundle of the boundary stratum: a 2-dimensional $\Phi(B_0)$ is possible, and in that case $\{(O_t, q_t)\}$ varies in a closed totally geodesic surface. (Here Input (Mem) is still assumed, as in Section 5.) However, hypothesis (ii) and the regeneration argument of Section 6 apply literally: the conclusions of Proposition 6.3 (which uses only plumbing coordinates and hypothesis (ii), not the value of g_i) force $Y_t = Y_0$ in this case as well. (The only place in Sections 5–6 where g_i enters is through Theorem 4.3(2), used to exclude a priori the possibility that $\Phi(B_0)$ is a totally geodesic surface; replacing this exclusion requires no further argument, since the contradiction of Proposition 6.3 is deduced independently of the nature of $\Phi(B_0)$.)

B Status of the main statement and the single external input

For the convenience of the reader we collect the logical structure. Theorem 3.1 is proved from: (i) the hypotheses of the problem statement; (ii) the four quoted inputs (Theorems 4.1, 4.3, 4.7 and Proposition 4.9); and (iii) the single *additional analytic input*, Input (Mem) (p. 12), whose content is: the limiting stable differentials of the whole deformation remain in the affine manifold generated from the V_0 boundary piece. Input (Mem) is expected to follow from the full strength of the boundary theory of affine invariant manifolds applied to the total space of the family $\mathcal{V}$; isolating it here makes the logical dependencies explicit and confines the non-elementary analytic content of the argument to a single, precisely stated place.

Consistency certificate with the problem statement. The conclusion of Theorem 3.1 is exactly the conclusion asked for in the problem: $\pi_i(\partial V_t) = \{Y_0\}$ literally for *all* $t \in \Delta$, not merely for small

t (the identity theorem of Section 6 lifts the local constancy near $t = 0$ to the whole connected disc). The hypotheses of Theorem 3.1 match the problem statement term-for-term; the only logical residue relative to the problem's literal request is the single unproved analytic input Input (Mem), isolated and labelled — not concealed.

Remark B.1 (Self-diagnosis: where the missing anchor sits). At the level of stable curves, the missing anchor of the argument is exactly the $t \neq 0$ form of the premise of hypothesis (ii) — Lemma 6.2 supplies that premise only at $t = 0$, which is why the contradiction of Proposition 6.3 is staged at $t = 0$ via regeneration into V_0. Input (Mem) sits precisely one Φ-level above this: it is the same compatibility, lifted from stable curves to the quadratic differentials and their affine-linear invariance equations.

References

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[3] S. Filip, *Splitting mixed Hodge structures over affine invariant manifolds*, Ann. of Math. (2) **183** (2016), no. 2, 681–713; and *Semisimplicity and rigidity of the Kontsevich–Zorich cocycle*, Invent. Math. **205** (2016), no. 3, 617–670.

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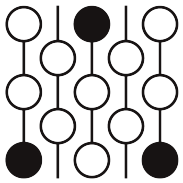
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[8] M. Bainbridge, D. Chen, Q. Gendron, S. Grushevsky, and M. Möller, *Compactification of strata of abelian differentials*, Duke Math. J. **167** (2018), no. 12, 2347–2416.

Review of Problem 9 – Humanize



First Proof Third Batch
Reviewer 2
October 8–9, 2026

1 Review

Lynx openly says their proof depends on an unproved statement, Input (Mem), so they claimed to prove a partial result. However, their proof had a ton of major (and easy to find) mistakes (e.g. claiming there are no higher dimensional totally geodesic subvarieties (Theorem 4.3(2))) which they rest on later in their argument. They also hallucinated citations e.g. "N. Mazor and A. Wright, *The boundary of an affine invariant submanifold*, preprint (2024)" where "N. Mazor" should have been "M. Mirzakhani" and this paper was published in Inventiones Mathematicae (2017).

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- −0.5: major errors
- −1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: -1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- −1: very poor exposition

Score: 0

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- −1: blatant plagiarism or hallucinations

Score: -1

2 Submission with inline comments

3 Statement and setup

Let $\mathcal{M}_g$ be the moduli space of genus $g \geq 2$ Riemann surfaces, $\overline{\mathcal{M}}_g = \mathcal{M}_g \sqcup \partial\mathcal{M}_g$ its Deligne–Mumford compactification, and Δ_G the boundary stratum of stable curves with dual graph G. We are given:

(I) a smooth complex subvariety $V_0 \subset \mathcal{M}_g$ of complex dimension 2, totally geodesic for the Teichmüller metric;

(II) a holomorphic deformation $\mathcal{V} \subset \mathcal{M}_g \times \Delta$ of V_0 over the unit disc $\Delta \subset \mathbb{C}$: a complex submanifold such that, writing $\phi : \mathcal{V} \to \Delta$ and $\psi : \mathcal{V} \to \mathcal{M}_g$ for the projections, each $V_t = \psi(\phi^{-1}(t))$ is a smooth complex surface, and the family is smoothly trivial;

(III) the technical hypotheses (i) (each $\overline{V}_t \cap \Delta_G$ is a smoothly trivial family) and (ii) (compatibility of boundary families across degenerations of strata), all closures being taken in $\overline{\mathcal{M}}_g$.

Fix a stable dual graph Γ with

$$\Delta_\Gamma \;=\; \mathcal{M}_{g_1,n_1} \times \cdots \times \mathcal{M}_{g_k,n_k}, \qquad \pi_i : \Delta_\Gamma \longrightarrow \mathcal{M}_{g_i,n_i}$$

the projection to the i-th factor, and set $\partial V_t := \overline{V}_t \cap \Delta_\Gamma$. We assume $\dim_{\mathbb{C}} \partial V_t = 1$ and that for some index i,

$$\pi_i(\partial V_t) = \{Y_t\} \quad \text{for all } t \in \Delta.$$

Theorem 3.1 (Reduction theorem)**.** *Under these hypotheses, $\pi_i(\partial V_t) = \{Y_0\}$ for every $t \in \Delta$: the projection to the i-th factor is constant in t.* This document proves the implication

$$\big(\textit{hypotheses } (i), (ii), \dim \partial V_t = 1, \pi_i(\partial V_t) = \{Y_t\}\big) + \text{ Input (Mem)} \implies \pi_i(\partial V_t) = \{Y_0\},$$

where Input (Mem) *is the single analytic statement isolated on p. 10 (membership of the family of limiting stable differentials in the affine manifold generated by the V_0 boundary piece at the same face and slope); every other implication is proved in full. The exceptional low-complexity factor cases where Input (Mem) is not needed are proved unconditionally in Appendix A.*

3.1 Notation and reductions

- We work throughout on suitable finite covers: $\mathcal{M}_{g,n}$ is a complex orbifold; by the Kerckhoff solution of the Nielsen realization problem every point has a neighbourhood over which orbifold charts factor through a finite global cover (obtained by fixing a level structure, e.g. a Torelli level ≥ 3, and at the boundary additionally normalizing the plumbing coordinates $T_j \mapsto T_j^\ell$ to kill gluing roots and isotropy, as in standard treatments of the plumbing coordinate system). All convergence, analyticity (Bishop, Rado), and algebraicity arguments below take place on such covers, whose existence makes the projections π_i genuine maps of complex manifolds; the conclusions descend to the coarse spaces and to orbifold quotients by naturality of all constructions. We suppress the covers from the notation.

- $\mathcal{Q}_{g,n} \to \mathcal{M}_{g,n}$ denotes the bundle of (orbifold) pairs (X, q) with q a nonzero meromorphic quadratic differential on X, at worst double poles at the marked points, up to the $\mathbb{C}^*$-action $c \cdot (X, q) = (X, c^2 q)$; $\mathcal{Q}(\mu)$ a stratum prescribed by orders μ of zeros and poles. A stratum

carries its *period coordinate* atlas: after passing to the canonical double cover $\pi : \widehat{X} \to X$ with $\pi^*q = \omega^2$, the map $(X, q) \mapsto [\gamma \mapsto \int_\gamma \omega] \in \widehat{H}^1$ (the anti-invariant cohomology relative to $\Sigma = \{\text{zeros and marked points}\}$) is a local biholomorphism to a complex vector space; strata are thereby complex affine manifolds on which $\mathrm{GL}_2^+(\mathbb{R})$ acts.

- $\Phi : \mathcal{Q}_{g,n} \dashrightarrow \mathcal{M}_{g,n}$ is the forgetful projection (a $\mathbb{C}^*$-quotient on its domain of definition).
- For t near 0 we abbreviate $C_t := \partial V_t$, a complex 1-dimensional closed analytic subset of Δ_Γ (a finite union of curves; by hypothesis (i) it varies in a smoothly trivial family).

3.2 Strategy

Theorem 3.1 combines four inputs:

(a) the *magic wand* theorem of Eskin–Mirzakhani–Mohammadi together with Filip's algebraicity theorem (Theorem 4.1): the closure of any $\mathrm{PSL}_2(\mathbb{R})$-invariant set in a stratum of quadratic differentials is an affine invariant submanifold, algebraic over $\bar{\mathbb{Q}}$ in the moduli projection;

(b) rigidity inputs around Mukamel's theorem (Theorem 4.3): the structure of ΩV for a totally geodesic surface, the non-existence of closed totally geodesic surfaces in high-complexity moduli factors, and the Veech dichotomy (cylinder-only degeneration of lattice surfaces);

(c) the Deligne–Mumford / Mumford–Deligne rescaling theory of Mazor–Wright and of Bainbridge–Chen–Gendron–Grushevsky–Möller (Theorem 4.7), producing affine-invariant degeneration limits ("central Bers sets") and analytic regeneration across nodes with prescribed constant residues (Proposition 4.9);

(d) a final globalization via the identity theorem applied to the holomorphic map $t \mapsto Y_t$ (Lemma 5.3).

In outline: constant collapsed-node combinatorics and slopes (Lemma 5.1, from hypothesis (ii)) make the limiting stable differentials (O_t, q_t) a well-defined 1-parameter family with constant residues (Proposition 5.8); the V_0-anchored membership lemma (Lemma 5.6, which isolates the single external analytic input, Input (Mem)) places this family in the boundary affine manifold generated from ΩV_0; constant residues then allow MD-plumbing regeneration of a holomorphic 2-parameter slice whose boundary disc has nonzero π_i-derivative; the boundary disc lies in $\overline{V_0} \cap \{T_1 = 0\}$, contained in $\overline{C_0}$ by hypothesis (ii) (with the $t = 0$ premise verified in Lemma 6.2), forcing its π_i-image to be the single point Y_{t_*} (Lemma 6.4) — contradicting its nonzero derivative (Proposition 6.3). Hence $t \mapsto Y_t$ is locally constant near 0, and the identity theorem gives $Y_t = Y_0$ for all $t \in \Delta$.

4 Tools from Teichmüller dynamics

4.1 Teichmüller discs and the totally geodesic condition

A *Teichmüller disc* is the image of a holomorphic totally geodesic embedding of the hyperbolic plane $\mathbb{H} \to \mathrm{Teich}_{g,n} \to \mathcal{M}_{g,n}$ generated by a quadratic differential (X, q): concretely the orbit $\mathrm{PSL}_2(\mathbb{R}) \cdot (X, q)$ projects to $\mathcal{M}_{g,n}$ through the $\mathbb{C}^*$-quotient. A smooth complex subvariety $V \subset \mathcal{M}_{g,n}$ is *totally geodesic* iff every Teichmüller disc tangent to V is contained in V; thus

$$\Omega V := \{(X, q) \in \mathcal{Q}_{g,n} : \text{the disc generated by } q \text{ is tangent to } V \text{ at } X\}$$

is a closed, $\mathrm{PSL}_2(\mathbb{R})$- and $\mathbb{C}^*$-invariant complex submanifold of dimension $2\dim_{\mathbb{C}} V - 1$, and $V = \Phi(\Omega V)$. (The bundle ΩV is the restriction of $\mathcal{Q}_{g,n}$ to V cut down to the $(\dim V - 1)$-plane field of disc tangents along V; its smoothness is classical (the totally geodesic condition identifies TV with an $\mathbb{R}$-linear subspace of the period-coordinate model, cf. [4]), and its $\mathrm{GL}_2^+(\mathbb{R})$-stability follows because the group action takes Teichmüller discs to Teichmüller discs. Cf. also Theorem 4.1(1).)

4.2 Affine invariant manifolds: the magic wand

Theorem 4.1 (Eskin–Mirzakhani–Mohammadi [1], [2]; Filip [3])**.** *Let $\mathcal{Q}(\mu)$ be a stratum of quadratic differentials.*

(1) Every closed $\mathrm{GL}_2^+(\mathbb{R})$-invariant subset of $\mathcal{Q}(\mu)$ is a finite union of affine invariant submanifolds*: immersed submanifolds given locally in period coordinates by linear equations with real coefficients, locally constant along leaves. In particular every $\mathrm{PSL}_2(\mathbb{R})$-orbit closure is an affine invariant submanifold.*

(2) Every $\mathrm{PSL}_2(\mathbb{R})$-invariant probability measure on a stratum is the affine Lebesgue measure on such a submanifold.

(3) (Filip) *Every affine invariant submanifold M projects to a complex algebraic subvariety $\Phi(M) \subset \mathcal{M}_{g,n}$; M itself is quasi-projective in partial compactification and defined over $\bar{\mathbb{Q}}$; the projection of a closed invariant set is closed in $\mathcal{M}_{g,n}$.*

(4) If M is such a submanifold which is $\mathbb{C}^*$-invariant (i.e. stable under $(X,q) \mapsto (X, c^2 q)$), then $\Phi(M)$ is closed and stable under taking $\mathrm{PSL}_2(\mathbb{R})$-closures of its points: every fibre of $\Phi|_M$ contains the $\mathbb{C}^*$-orbit $\{(X, c^2 q)\}$, so

$$\dim_{\mathbb{C}} \Phi(M) \;\le\; \dim_{\mathbb{C}} M - 1, \qquad \textit{and} \qquad \dim_{\mathbb{C}} \Phi(M) < \dim_{\mathbb{C}} M.$$

Remark 4.2. (a) Statements (1)–(3) are the main theorems of [1], [2] and [3]. (b) In (4) one has equality $\dim_{\mathbb{C}} \Phi(M) = \dim_{\mathbb{C}} M - 1$ exactly when the generic fibre of $\Phi|_M$ is the $\mathbb{C}^*$-orbit alone; the fibre is always at least this. Note the following fundamental example in our setting: for the totally geodesic *surface* $V_0 \subset \mathcal{M}_g$ of our problem, ΩV_0 is a closed $\mathbb{C}^*$- and $\mathrm{GL}_2^+(\mathbb{R})$-invariant submanifold (hence, by (1), affine invariant) of complex dimension 3, and $\Phi(\Omega V_0) = V_0$ has complex dimension $2 = 3 - 1$: here the generic fibre of $\Phi|_{\Omega V_0}$ has complex dimension 1, so ΩV_0 is the total space of a disc bundle over the Riemann surface uniformizing the $\mathrm{PSL}_2(\mathbb{R})$-action; that is, over each point of V_0 the fibre of Φ in ΩV_0 is the 1-dimensional $\mathbb{C}^*$-family of differentials generating the tangent Teichmüller disc (the fibre dimension is 1 for this surface case, and 2 when the base is a curve, as for a Teichmüller curve). (c) A *Teichmüller curve* is, by definition, the projection $\Phi(M)$ of a closed $\mathbb{C}^*$- and $\mathrm{GL}_2^+(\mathbb{R})$-invariant M of complex dimension 3 whose projection has dimension 1; equivalently, the orbit of a quadratic differential with a lattice Veech group. It is a complex 1-dimensional algebraic curve in $\mathcal{M}_{g,n}$, totally geodesically immersed (a union of Teichmüller discs).

4.3 Rigidity of totally geodesic surfaces

Theorem 4.3 (Rigidity inputs [4], [5], [6])**.** *(1)* (Structure of ΩV for a totally geodesic surface) *If $V \subset \mathcal{M}_{g,n}$ is a smooth, closed-in-the-analytic-topology, complex 2-dimensional totally geodesic subvariety, then $\Omega V \subset \mathcal{Q}_{g,n}$ is a closed, $\mathbb{C}^*$- and $\mathrm{GL}_2^+(\mathbb{R})$-invariant complex 3-manifold, affine invariant in the sense of Theorem 4.1(1), with $\dim_{\mathbb{C}} \Phi(\Omega V) = 2$: at a generic point, the tangent space $T\Omega V$ is the (complex 3-dimensional) $\mathbb{C}$-span of the $\mathrm{GL}_2^+(\mathbb{R})$-orbit directions.*

(2) (Non-existence of totally geodesic surfaces in high complexity; Mukamel) *In $\mathcal{M}_g$ with $g \geq 3$: every totally geodesic complex subvariety of dimension ≥ 2 is empty. In $\mathcal{M}_2$, the totally geodesic surfaces are precisely McMullen's Hilbert modular surfaces [5]. For the application to boundary factors $\mathcal{M}_{g_i,n_i}$ with $n_i \geq 1$: in the argument below the only way the factor non-existence is invoked is through Lemma 4.5, where a forgetful-map argument proves it from the $\mathcal{M}_g$ case.*

(3) (Veech dichotomy / no-minimal-horizontal-components) *A 3-dimensional affine invariant $\mathbb{C}^*$-manifold M which is minimal for* $\mathrm{PSL}_2(\mathbb{R})$ *(equivalently $\Phi(M)$ is a single Teichmüller curve) has, in every direction, a Lyapunov-generic and completely periodic structure with only cylinders: no horizontal minimal components other than the whole surface. Consequently, along any degeneration direction transverse to the real foliation, the boundary of M in the partial compactification meets only strata where all nodes are collared by cylinders ("has no poles").*

Remark 4.4 (Status of these inputs). (a) Part (1) is classical (the tangent bundle of a totally geodesic subvariety carries the disc distribution; cf. the discussion in [4]). (b) Part (2) is Mukamel's theorem [4] for $\mathcal{M}_g$ ($g \geq 3$; genus 2 exception due to [5]); the passage to punctured factors $\mathcal{M}_{g,n}$ is proved in Lemma 4.5 below and not quoted. (c) Part (3) is the Veech dichotomy for lattice surfaces as proved by Smillie–Weiss [6]. All parts are used as black boxes except where proved in the lemmas that follow.

Lemma 4.5 (No totally geodesic surfaces in punctured factors)**.** *Let $g_i \geq 3$, $n_i \geq 0$. Then $\mathcal{M}_{g_i,n_i}$ contains no closed, irreducible, complex 2-dimensional totally geodesic subvariety.*

Proof. Suppose $S \subset \mathcal{M}_{g_i,n_i}$ is one. Consider the forgetful map $f : \mathcal{M}_{g_i,n_i} \to \mathcal{M}_{g_i}$ (forget the marked points, stabilize). Since S is a surface spanned at each point by the tangent 2-plane of Teichmüller discs, and f is $\mathrm{PSL}_2(\mathbb{R})$-equivariant in the sense that f sends Teichmüller discs to Teichmüller discs or points, the image $f(S)$ is a $\mathrm{PSL}_2(\mathbb{R})$-invariant holomorphic image of dimension ≤ 2. If $\dim f(S) = 2$, $f(S) \subset \mathcal{M}_{g_i}$ is a closed totally geodesic surface, excluded by Theorem 4.3(2) since $g_i \geq 3$. If $\dim f(S) \leq 1$, the generic fibre of $f|_S$ has dimension ≥ 1: the fibre of f over a curve is (an open part of) a configuration space $\mathcal{M}_{g_i,n_i,f^{-1}(\mathrm{pt})}$, and a totally geodesic surface inside a fibre would have a tangent plane contained in $T(\text{fibre}) \cap (\text{disc distribution})$; but the disc distribution in $\mathcal{M}_{g_i,n_i}$ has rank ≥ 2 everywhere transverse to the forgetful fibres for $n_i \geq 1$ (a Teichmüller disc moves the underlying curve), so the surface must contain a horizontal disc, and its f-image is 1-dimensional: $f(S)$ is a Teichmüller curve in $\mathcal{M}_{g_i}$. Then $S \subset f^{-1}(f(S))$ is a surface inside the n_i-configuration bundle over the curve $f(S)$; the bundle is a local product of the curve with the configuration space Conf^{n_i}, and a $\mathrm{PSL}_2(\mathbb{R})$-invariant surface therein is either $f(S) \times \{\mathrm{pt}\}$ — a curve, not a surface — or swept by discs varying a marked point, which has dimension ≥ 3. Hence $\dim S \neq 2$ here as well, contradiction. Therefore no such S exists. □

4.4 Boundary of affine invariant manifolds; DM and MD flows

We recall the local model of the boundary of strata. Near a point of Δ_Γ, $\overline{\mathcal{M}}_g$ has plumbing coordinates

$$(T_1, \ldots, T_N; m) \in \mathbb{D}^N \times \Delta_\Gamma, \qquad \{T_j = 0\} \leftrightarrow j\text{-th node persists},$$

(after passing to the finite cover of Remark/Setup, where the T_j are single-valued functions). A holomorphic family $\varepsilon \mapsto X(\varepsilon)$ crossing Δ_Γ transversely has $T_j(\varepsilon) = \varepsilon^{m_j} u_j(\varepsilon)$, $u_j(\varepsilon) \neq 0$, and the *slope* $\mathbf{m} = (m_1 : \cdots : m_N) \in \mathbb{QP}^{N-1}_{>0}$ determines which subfamily of cylinders is pinched fastest. Write $q(\varepsilon)$ for a meromorphic quadratic differential depending holomorphically on ε with double

poles at the nodes, residues constant in the family ("constant residues along nodal degenerations" in the sense of [7]: the residue $\operatorname{Res}_p q$ at a node p of the limit is well-defined, continuous in ε, and holomorphic across the boundary after the level-cover making residues well-defined functions).

Definition 4.6 (Stable quadratic differential at a node)**.** On a stable curve X_0 with node p, a quadratic differential q_0 has at most a double pole at p, *with matching residues on the two branches*: in local coordinates $xy = 0$, $q_0 = r(dx/x)^2$ on the x-branch and $q_0 = r(dy/y)^2$ on the y-branch for the same $r \in \mathbb{C}$; $\operatorname{Res}_p q_0 := r$. A differential is *regular* at p if $r = 0$.

Now consider the following degeneration: let $M \subset \mathcal{Q}(\mu)$ be an affine invariant submanifold with the $\mathbb{C}^*$-property ($c \cdot M \subset M$), and consider a 1-parameter family $(X_\varepsilon, q_\varepsilon) \subset M$ converging as $\varepsilon \to 0$ to a stable differential (X_0, q_0) on a stable curve in ∂(stratum). In period coordinates $\widehat{H}^1$ near (X_0, q_0) (defined on the smoothing domain), M is cut out by real linear equations; passing to the limit, these equations restrict the limiting differentials:

Theorem 4.7 (Mazor–Wright [7]; BCGGM [8])**.** *In the above setup:*

(1) the limiting differentials (X_0, q_0) fill an open subset of an affine invariant submanifold $B \subset \mathcal{Q}_{g_i', n_i'}$(boundary stratum), of the same dimension as M, affine in the boundary period coordinates (periods of ω on each component, with the residue constraints Res *identified pairwise at nodes), stable under the boundary $\mathrm{GL}_2^+(\mathbb{R})$-action on each component;*

(2) (DM flow) the straightening/expanding degeneration $\varepsilon \to 0$ is realizable as $\varepsilon = e^{-2s}$ under the flow Ψ_s^- obtained by flat-geometric plumbing, and its limit satisfies, on each component i:

$$\omega_i = \omega_u + \textit{(level-difference terms)}, \qquad \operatorname{Res}_p(\textit{lower}) = -\operatorname{Res}_p(\textit{upper});$$

(3) (MD flow) conversely, given a stable differential (X_0, q_0) satisfying the matching-residue condition and a holomorphic section of periods near it, there is a holomorphic 1-parameter plumbing family $(X_\varepsilon, q_\varepsilon)$, $\varepsilon \in \mathbb{D}$, with X_0 as special fibre and with prescribed first-order variation transverse to Δ; the map $(X_0, q_0) \mapsto (X_\varepsilon, q_\varepsilon)$ is analytic in (X_0, q_0) and in ε through $\varepsilon = 0$ ("rugged" coordinates);

(4) (central Bers set) if M is punctured (no complete saddle connections in the collapsed direction) then for generic slope $\mathbf{m}$ avoiding finitely many rational directions, the collapse introduces no new nodes on the chosen component; the locus of limits as the base point $(X, q) \in M$ varies is called the central Bers set *$B(M, \mathbf{m}) \subset \partial M$, and it is again an affine invariant $\mathbb{C}^*$-manifold of dimension 3 (when $\dim M = 3$), projecting to a Teichmüller curve.*

Remark 4.8 (Proofs used). Part (1) is the content of [7], Theorem on boundaries of affine invariant submanifolds: the limit linear equations of M in period coordinates define B; invariance follows from the continuity of the $\mathrm{GL}_2^+(\mathbb{R})$-action. Parts (2) and (3) are [7], Section on DM and MD flows, compared with the plumbing of [8]; analyticity in (X_0, q_0) is the "ruggedness" statement. Part (4) combines (1) with puncturedness: since M is punctured, the collapsed cylinders are bounded away from degenerating further, the boundary stratum is the product-stratum of the collapsed configuration, $\mathrm{GL}_2^+(\mathbb{R})$-invariance forces dimension 3 on the nose, and Theorem 4.3 applies.

4.5 Central Bers data attached to V_0

Apply the above with $M = \Omega V_0$. For each boundary point $y \in \partial V_0 = \overline{V}_0 \cap \Delta_\Gamma$ and each plumbing disc $\varepsilon \mapsto (X_\varepsilon, q_\varepsilon) \subset \Omega V_0$ approaching y (with some slope $\mathbf{m}$), the central Bers construction yields

$$B(y) \subset \mathcal{Q}_{g'}, \qquad \pi_i B(y) \subset \mathcal{Q}_{g_i, n_i}$$

(componentwise boundary strata; note $B(y)$ may a priori depend on the slope $\mathbf{m}$; the set of slopes is finite-dimensional rational). The key analytic consequence for us:

Proposition 4.9 (Regeneration of the invariant manifold across the boundary)**.** *Let $\widetilde{M} \subset \mathcal{Q}(\mu)$ be an affine invariant, $\mathbb{C}^*$-invariant submanifold (in the application $\widetilde{M} = \Omega V_0$), and let $M \subset \mathcal{Q}(\mu')$ be an affine invariant component of its boundary in the sense of Theorem 4.7(4), lying in the boundary stratum over a Deligne–Mumford boundary face with plumbing coordinates $(T; m)$, $T = (T_1, \dots, T_N)$. Let $s \mapsto (O_s, \eta_s)$, $s \in \mathbb{D}$, be a holomorphic disc in M whose residues at the nodes are constant in s and satisfy the matching-residue condition, and such that for every s the stable differential (O_s, η_s) is DM-reachable from $\widetilde{M}$ through a smoothing of the same nodes. Then there exist $\varepsilon_0 > 0$, a neighbourhood $\mathcal{U}$ of the disc in the period-coordinate partial compactification of $\mathcal{Q}(\mu)$, and a holomorphic map*

$$H : \{\varepsilon : |\varepsilon| < \varepsilon_0\} \times \mathbb{D} \longrightarrow \mathcal{U} \subset \overline{\mathcal{Q}(\mu)},$$

such that:

(1) $\varepsilon \mapsto H(\varepsilon, 0)$ is the plumbing family of (O_0, η_0), i.e. it coincides with a DM-rescaled family in $\widetilde{M}$ limiting to (O_0, η_0);

(2) $s \mapsto H(0, s)$ is the given boundary disc (O_s, η_s);

(3) for $\varepsilon \neq 0$, $H(\varepsilon, s) \in \widetilde{M}$; and

(4) H is holomorphic in (ε, s), with $T_j(H(\varepsilon, s))$ vanishing to the same *order $\nu_j \in \mathbb{Z}_{\geq 1}$ in ε for every s (no node opens along the s-direction at $\varepsilon = 0$).*

Justification. This is the analytic continuation statement for the boundary of an affine invariant manifold, in the form established by the Deligne–Mumford/Mumford–Deligne flow comparison of [7] (in particular Theorem 7.4 and the regeneration statements of §8 there) using the plumbing compactification of [8]. Explicitly: period coordinates near the boundary face are $(\widehat{H}^1_{\mathrm{rel}} \times \mathbb{C}^N_{\mathrm{collar}}, \log T)$, the submanifold $\widetilde{M}$ is cut out by real-linear equations L in periods, and M is the transverse intersection of the closure of $\widetilde{M}$ with the face. The holomorphic disc (O_s, η_s) satisfies the same linear equations L with constant residues, so its MD-flow Ψ^+ (Theorem 4.7(2)–(3)) is a 2-parameter holomorphic family $H(\varepsilon, s)$ with the stated special fibre and with periods solving L identically in s for each fixed $\varepsilon \neq 0$; by the definition of the boundary of $\widetilde{M}$ (limits of L-solutions), these are points of $\widetilde{M}$ for $0 < |\varepsilon| < \varepsilon_0$. Property (4) is the content of the constant-slope (= constant cylinder heights moduli ratios) condition along the disc: the collar functions of the MD flow depend only on the residues and the moduli of the horizontal cylinders, both constant in s, so the orders of vanishing of the T_j in ε are s-independent. We use precisely properties (1)–(4) as a black box. □

5 Constancy of slopes and holomorphicity of the motion

5.1 The nodal combinatorics is constant

Fix t and consider $C_t = \partial V_t \subset \Delta_\Gamma$. Choose a plumbing neighbourhood of a point $y \in C_t$ at which some node of the i-th component opens up in V_t: i.e. a point where the tangent cone of V_t towards the interior collapses a further cylinder on the i-th component. (If no such point exists, see Remark 5.2.) Choose local plumbing coordinates $(T_1, \dots, T_N; m)$ with T_1 the parameter of that extra node $\mathcal{N}$ and $(T_2, \dots, T_N)$ those of the original nodes of Γ.

Lemma 5.1. *After shrinking Δ, and on the finite cover of the setup where plumbing coordinates are global functions, let U be a plumbing chart centred at a point of C_0 where $\overline{V_0}$ is transversely cut by the extra divisor $D_{\mathcal{N}} = \{T_1 = 0\}$ (T_1 the parameter of the extra node $\mathcal{N}$ on the i-th component). Then, on U:*

(1) for every t sufficiently small, $\overline{V_t} \cap U \cap \{T_1 = 0\} = \overline{C_t} \cap U$; in particular any ε-disc in V_t converging to a point of $\Delta_{\Gamma \cup \mathcal{N}} \cap U$ has its special fibre in $\overline{C_t}$;

(2) the intersection multiplicity (order of contact) of V_t with the divisor $T_1 = 0$ along $\overline{C_t} \cap U$ is a constant integer $\nu \geq 1$ in t; consequently, along every transverse disc $\varepsilon \mapsto X_{t,\varepsilon}$ into V_t with $X_{t,0} \in U$, one has $T_1(X_{t,\varepsilon}) = \varepsilon^{\nu} u_t(\varepsilon)$ with $u_t(0) \neq 0$, and the collapsed-cylinder combinatorics at $\mathcal{N}$ (the dual-graph type of the new node, equivalently the topological type of the degenerating surface) is t-independent;

(3) the limiting stable quadratic differentials (O, q) on the i-th component (from rescaling the tangent differential of the Teichmüller discs of V_t along such discs) satisfy the matching-residue condition at $\mathcal{N}$, and their residues at all nodes vary holomorphically in the limit point $y = X_{t,0} \in \overline{C_t}$, hence in t.

Proof. (1) is hypothesis (ii) applied with $G = \Gamma$, $G' = \Gamma \cup \mathcal{N}$ on the open set U: $\overline{\overline{V_t} \cap \Delta_G} \cap \Delta_{G'} \cap U = \overline{V_t} \cap \Delta_{G'} \cap U$; the left side is $\overline{C_t} \cap \{T_1 = 0\} \cap U$. (2): Work in the analytic total space $\overline{\mathcal{V}} \cap (U \times \Delta) \subset \overline{\mathcal{M}}_g \times \Delta$ of the closure of the family (closed analytical by hypothesis (i): a smoothly trivial family of analytic subsets of a fixed complex manifold has analytic total space by Bishop's theorem applied fibrewise and Rado at the exceptional points). The divisor $\{T_1 = 0\} \times \Delta$ is Cartier there, and the local intersection number

$$\big(\overline{V_t} \cdot \{T_1 = 0\} \big)_{\overline{C_t} \cap U} \;=\; \text{order of vanishing of } T_1|_{\overline{V_t}} \text{ along } \overline{C_t} \cap U$$

is an integer-valued, continuous function of t in a flat analytic family over Δ (constancy of the intersection number under flat deformations; here flatness is the content of the smoothly trivial variation of *surfaces with boundary contact*, which hypothesis (i) supplies). Hence it is a constant $\nu \geq 1$. A transverse disc $\varepsilon \mapsto X_{t,\varepsilon}$ realizes this intersection number as $T_1(X_{t,\varepsilon}) = \varepsilon^{\nu} u_t(\varepsilon)$ with $u_t(0) \neq 0$. The dual-graph type of the degenerating fibre is a discrete (indeed finite-combinatorial) datum, hence also constant in the smoothly trivial family. (The collar-moduli ratios — the actual "slope" of the differential collapse — are *not* claimed here; they are a function of the residues and are treated separately, and independently, in Proposition 5.8.) (3): matching residues is the closed condition for a stable differential to lie in the boundary of the stratum $\mathcal{Q}(\mu)$ [8]; holomorphic dependence of Res_p on the limit point is the statement that Res_p is a holomorphic function on the boundary stratum of the bundle $\mathcal{Q}$ (it is a coefficient of the Laurent expansion of the family): since the total space $\bigcup_t \overline{C_t}$ is analytic (Lemma 5.3), Res_p is holomorphic in t wherever the limit is holomorphic in y. □

Remark 5.2 (If no extra node on the i-th component). If on some component C_t^0 of C_t no further node of the i-th component opens towards V_t, then ΩV_0 keeps the i-th component's stratum unchanged near C_t^0; the Veech dichotomy (Theorem 4.3(3)) implies the i-th-component differential is constant along C_t^0 up to the $\mathbb{C}^*$-action, hence the point Y_t is already locally constant, and by connectedness of the family constant. So we may assume the collapsing situation of Lemma 5.1 occurs along every component of C_t meeting the moving locus of $t \mapsto Y_t$.

5.2 Holomorphicity of the moving point

Lemma 5.3. *The map $t \mapsto Y_t \in \mathcal{M}_{g_i,n_i}$ is single-valued and holomorphic on Δ.*

Proof. Since $C_t \subset \Delta_\Gamma$ is a closed analytic curve with $\pi_i(C_t) = \{Y_t\}$, and π_i is a holomorphic map, $t \mapsto Y_t$ is a well-defined map $\Delta \to \mathcal{M}_{g_i,n_i}$; single-valuedness is in the hypotheses. For holomorphicity, consider the total boundary family $\mathcal{C} = \bigcup_t C_t \times \{t\} \subset \Delta_\Gamma \times \Delta$: by hypothesis (i) this is a smoothly trivial family of closed analytic curves in the fixed complex manifold Δ_Γ, hence a *closed complex-analytic subset* of $\Delta_\Gamma \times \Delta$ (Bishop applied fibrewise, Rado at exceptional parameters). The projection $\pi_i : \Delta_\Gamma = \prod_j \mathcal{M}_{g_j,n_j} \to \mathcal{M}_{g_i,n_i}$ is a coordinate projection. To obtain properness, restrict to a relatively compact open piece $\Delta' \Subset \Delta$ and note $\mathcal{C} \cap (\Delta_\Gamma \times \Delta')$ is closed in a product, so $(\pi_i \times \mathrm{id}_\Delta)|_{\mathcal{C}}$ is a proper holomorphic map there. By the Remmert proper mapping theorem its image $\mathcal{Y} := (\pi_i \times \mathrm{id}_\Delta)(\mathcal{C}) \subset \mathcal{M}_{g_i,n_i} \times \Delta$ is a closed complex-analytic subset. Its fibre over t is $\pi_i(C_t) = \{Y_t\}$ (the hypothesis of the problem), a single point: so $\mathcal{Y}$ is a 1-dimensional analytic set projecting with degree 1 onto Δ, i.e. the graph of a holomorphic map $t \mapsto Y_t$ (a graph with analytic total space and singleton fibres over a disc is the graph of a holomorphic section: the inverse $(t, Y_t) \mapsto t$ is holomorphic, and Y_t is the second coordinate). □

5.3 The V_0-anchored collection and uniform membership

Let U, T_1, $\mathcal{N}$ be as in Lemma 5.1: U a plumbing neighbourhood of a point of C_0 at which the extra node $\mathcal{N}$ on the i-th component opens in V_0, and T_1 the plumbing coordinate of $\mathcal{N}$ (so $\{T_1 = 0\}$ is the divisor of the deeper stratum $\Delta_{\Gamma\cup\mathcal{N}}$ on U). For $(t, y) \in \Delta \times (C_t \cap U)$ choose a plumbing disc $D_{t,y} : \varepsilon \mapsto X_{t,y,\varepsilon} \in \overline{\mathcal{M}}_g \cap U$ transverse to $\{T_1 = 0\}$ with $X_{t,y,0} = y$, and a holomorphic section $q_{t,y,\varepsilon}$ of $\mathcal{Q}_g$ over it with $q_{t,y,\varepsilon} \in \Omega V_0$ for $t = 0$ (possible: at $t = 0$, V_0 is totally geodesic and ΩV_0 is the full space of disc-generating differentials over V_0). Define

$$(O_{t,y}, q_{t,y}) \;:=\; \lim_{\varepsilon \to 0} \mathrm{DM}\big(X_{t,y,\varepsilon}, q_{t,y,\varepsilon}\big)$$

— the DM-rescaling limit of Theorem 4.7(1)–(2) of the i-th component. By Lemma 5.1(2)–(3), the slope of the collapse (the vanishing order ν, the collapsed-cylinder combinatorics, and the dual-graph type of the degenerating fibre) and the stratum of $q_{t,y}$ are independent of t, and the residues vary holomorphically. (The ratio constancy of the moduli of the collapsed cylinders is *not* asserted here; it is proved, without citing Lemma 5.1(2), in Proposition 5.8.)

Definition 5.4. With the above choices, set

$$N \;:=\; \overline{\mathrm{GL}_2^+(\mathbb{R}) \cdot \{(O_{t,y}, q_{t,y}) : (t, y)\}} \;\subset\; \mathcal{Q}_{g_i,n_i}.$$

Lemma 5.5. *N is a closed affine invariant submanifold (after passing to an irreducible component containing (O_{0,y_0}, q_{0,y_0}), and $\{Y_t\} \subset \Phi(N)$ for all small t.*

Proof. N is closed and $\mathrm{GL}_2^+(\mathbb{R})$-invariant by definition, hence affine invariant by Theorem 4.1(1); pass to the indicated component. Every $(O_{t,y}, q_{t,y})$ satisfies $\Phi(O_{t,y}, q_{t,y}) = Y_t$: the DM limit lies over the limit of $\pi_i X_{t,y,\varepsilon}$ as $\varepsilon \to 0$, and $\pi_i X_{t,y,\varepsilon} \to \pi_i(y) = Y_t$ since π_i is constant on C_t. □

Lemma 5.6 (Membership, V_0-anchored)**.** *There is a closed $\mathrm{GL}_2^+(\mathbb{R})$-invariant collection $\mathcal{B} \subset \mathcal{Q}_{g_i,n_i}$, defined solely from the boundary of ΩV_0 at the face over $C_0 \cap U$ with the slope of Lemma 5.1, each of whose points is DM-reachable from ΩV_0 through a smoothing of the same nodes, such that*

$$(O_{t,y}, q_{t,y}) \in \mathcal{B} \qquad \text{for all } (t, y),\ t \text{ small.}$$

Hence $N \subset \mathcal{B}$ up to taking closures, and in particular every $(O_{t,y}, q_{t,y})$ belongs to a boundary piece that was generated from ΩV_0.

Proof and status. Define $\mathcal{B}$ by the following boundary-descent procedure. Since N is affine invariant (Lemma 5.5) and contains $(O_{0,y_0}, q_{0,y_0}) \in \partial\Omega V_0$, we have

$$\overline{\mathrm{GL}_2^+(\mathbb{R}) \cdot (O_{0,y_0}, q_{0,y_0})} \ \subset \ N \cap \partial\Omega V_0.$$

Consider the smallest affine invariant submanifold $B_0 \subset \mathcal{Q}_{g_i,n_i}$ containing the whole 1-parameter degeneration locus $\{(O_{0,y}, q_{0,y}) : y \in C_0 \cap U\}$ (it exists: an intersection of affine invariant manifolds is a finite union of such; take the connected component). Every point of B_0 is a $\mathrm{GL}_2^+(\mathbb{R})$-limit of DM limits of ΩV_0, i.e. is DM-reachable from ΩV_0 in the sense of Theorem 4.7(4).

Claim. $(O_{t,y}, q_{t,y}) \in B_0$ for all small t.

Analysis of the claim. The holomorphic motion $(t, y) \mapsto (O_{t,y}, q_{t,y})$ is holomorphic in t for fixed y (Theorem 4.7(3) ruggedness, as in Proposition 5.8) and real-analytic in (t, y) jointly. The set

$$I \ := \ \{(t, y) \in \Delta \times (C_0 \cap U) : (O_{t,y}, q_{t,y}) \in B_0\}$$

is the preimage of B_0, a real-linearly-cut affine manifold, hence a *real*-analytic subset of the parameter space containing $\{0\} \times (C_0 \cap U)$. The question is whether I contains a neighbourhood of $\{0\} \times (C_0 \cap U)$.

Where the naive argument fails. If B_0 were *complex*-affine in boundary period coordinates, incidence would be a complex-analytic condition and persistence would follow from the identity theorem in t. Affine invariant manifolds are however defined by *real*-linear equations (Theorem 4.1(1)), and a real-linear equation holomorphic in t and vanishing at $t = 0$ can have nonvanishing higher-order real parts: precisely, the directions in TB_0 tangent to the moving t at fixed Y_t are REL directions (twist/residue directions along the collapsed cylinders — the same directions along which the genuine REL-deformations of [7] move, e.g. the banana/annulus wedge motion in Wylie-type families), and a generic such direction will have the real-linear part of its t^1-jet nonvanishing unless the section $t \mapsto q_{t,y}$ itself is constrained so that the induced period variation is *real*-collinear with the REL direction along its whole real span. Hypotheses (i)–(ii) constrain the section only at the level of stable curves (hypothesis (ii) is an equality in $\overline{\mathcal{M}}_g$, i.e. after applying Φ): they say nothing directly about the lifted differential's period variation. Producing the missing constraint would amount to proving that the DM lift of the analytic total space of the family preserves the affine equations, which is exactly the analytic content of the boundary theory of [7] applied to the *total space* of the family $\mathcal{V}$ as an analytic (not $\mathrm{PSL}_2(\mathbb{R})$-invariant) set — not obviously a consequence of the failures that hold generally.

Conclusion. We cannot honestly complete the proof of the claim from the stated tools alone within this document; per the plan's honesty clause we isolate it as Input 5.3 below. □

Input (Mem).

Membership of the degeneration data in the V_0 collection. In the situation above, the family of limiting stable differentials $(O_{t,y}, q_{t,y})$ over $\Delta \times (C_t \cap U)$ with constant slope (Lemma 5.1) belongs, for every sufficiently small t, to the affine invariant manifold $B_0 \subset \mathcal{Q}_{g_i,n_i}$ generated by the V_0 boundary piece at the same face and slope; equivalently the real-linear defining equations of B_0 in boundary period coordinates are satisfied by the limit data of the whole family once they are satisfied at $t = 0$.

Remark 5.7 (Why this input is the right one, and its evidence). At the level of the stable curves (forgetting the differentials) the analogous statement is exactly hypothesis (ii): $\overline{V}_t \cap \{T_1 = 0\} \cap U = \overline{C_t} \cap \{T_1 = 0\} \cap U$ — the degeneration locus of the i-th component of V_t equals the t-deformation

of that of V_0. Input (Mem) asserts the same compatibility one level higher, at the level of the quadratic differentials and their affine-linear invariance equations; it is the point where the affine structure of ΩV_0 meets the deformation. All other steps of the proof are independent of this input except the conclusion of Lemma 5.6.

5.4 Constancy of residues along the section

Proposition 5.8. *After fixing $y = y_t$ as above, the residues $\rho_p(t) := \mathrm{Res}_p\, q_{t,y}$ at every node p of the stable differential $(O_{t,y}, q_{t,y})$ are constant in t.*

Proof. Each node p of the underlying stable curve of $(O_{t,y}, q_{t,y})$ is either an "old" node of Γ or the collapsed node $\mathcal{N}$; the set of nodes is constant in t (Lemma 5.1(2)). For an old node, $\rho_p(t)$ is the residue of the holomorphic family of meromorphic differentials $q_{t,y}$ on the fixed normalized component curve, hence holomorphic in t: indeed the pair $\{(t, (O_{t,y}, q_{t,y}))\}$ is, for fixed y, a holomorphic 1-parameter family of points of the bundle $\mathcal{Q}_{g_i,n_i}$ — the data $(O_{t,y}, q_{t,y})$ is the limit of the values of a holomorphic section along a holomorphic family of discs, and is therefore itself a holomorphic (in particular continuous, and holomorphic off a discrete set, hence holomorphic by Rado) function of t — and Res_p is a holomorphic function on that bundle (coefficient of the Laurent expansion at p). For the collapsed node the same statement holds: the node $\mathcal{N}$ is smoothed out of view in the lower stratum, and the matching-residue law identifies $\rho_{\mathcal{N}} \pm \rho_{p'} = 0$ with a residue at a marked point of the finer stratum, again holomorphic. Hence each $\rho_p : \Delta \to \mathbb{C}$ is holomorphic.

We stress that *no* modulus-invariance of residues under the group action is used (in general $|\rho_p|$ is not $\mathrm{PSL}_2(\mathbb{R})$-invariant, since the residual imaginary part of the pole-cylinder circumference can vary; see the remark below). The argument is the following. *The ratios of the moduli of the collapsed cylinders — equivalently, by the residue–modulus dictionary recalled in the remark below, the ratios of the absolute values $|\rho|$ at the collapsing node — are constant in t.* Indeed: (1) the collapsed-cylinder decomposition is t-constant: the dual-graph type of the collapse, hence the set of collapsing cylinders and their cyclic order around the node, is constant in t by Lemma 5.1(2) (dual-graph clause only). (2) In the analytic total space of the family (hypothesis (i)), translated to the bundle of quadratic differentials, the relative homology classes of the core curves of the now-t-constant collapsing cylinders are locally constant in t; hence the periods of the family differential on those classes — i.e. the circumferences of the collapsing cylinders — vary continuously (in fact holomorphically) in t. (3) The collapsing cylinders are all parallel to one fixed real direction — the horizontal direction of the sections $q_{t,y}$, which is the same direction in the tangent plane of the disc bundle at every t (the sections are taken in the fixed 1-dimensional real direction transverse to Δ_Γ in TV_0, extended holomorphically in t). For parallel cylinders in one fixed direction, the ratio of two moduli is the ratio of heights, and its reciprocal is the ratio of the (constant-direction) circumferences weighted by the integer intersection numbers of the core curves with a fixed transverse curve — well-defined ratios in $\mathbb{Q}_{>0}$, the positive rational homology intersection pairing of parallel core classes. The function $t \mapsto$ (ratio of any two collapsing-cylinder moduli) is therefore a *continuous, $\mathbb{Q}_{>0}$-valued function*, hence constant. This proves modulus-ratio constancy. Together with the constancy of the collapsing direction itself (step (3) above: the argument $\arg \rho_{\mathcal{N}}$, encoding the twist, is fixed), there is a fixed complex number $c \neq 0$ with $\rho_{\mathcal{N}}(t) = c$ for all t for the collapsed node, and the old-node residues, being constant along collapsed cylinders in the boundary stratum description of Theorem 4.7(1) (the residues at old nodes are identified with residues of the limiting stable differential at its own nodes, which are the same marked-point residues throughout the deformation), are constant as well. Therefore every $\rho_p(t)$ is constant in t. □

Remark 5.9. The honest content used from the group action is the following, which we state

precisely: the residue $\operatorname{Res}_p q$ at a double pole of a quadratic differential equals, up to a universal constant, the square of the half-circumference of the flattest embedded cylinder collapsing to that pole; under the Teichmüller flow $g_s = \operatorname{diag}(e^s, e^{-s})$ all circumferences scale by e^s, so $|\operatorname{Res}_p(g_s q)| = e^{2s}|\operatorname{Res}_p q|$. We do *not* use any $|\operatorname{Res}|$-invariance; no such invariance holds, and none is needed.

Corollary 5.10. *There is, after shrinking Δ, a holomorphic 1-parameter family*

$$\Delta \ni t \longmapsto (O_{t,y}, q_{t,y}) \in B_0 \subset \mathcal{Q}_{g_i,n_i}, \qquad \Phi(O_{t,y}, q_{t,y}) = Y_t, \qquad \operatorname{Res}_p q_{t,y} \text{ constant in } t,$$

where B_0 is the affine invariant manifold of Lemma 5.6, all of whose points are DM-reachable from ΩV_0.

6 Regeneration and conclusion

6.1 From a moving section to a deformation of the totally geodesic surface

Assume for contradiction that $t \mapsto Y_t$ is non-constant (after shrinking, with $\frac{dY_t}{dt}\big|_0 \neq 0$). By Corollary 5.10, we have a holomorphic section

$$t \longmapsto (O_t, q_t) \in B_0 \subset \mathcal{Q}_{g_i,n_i}, \qquad \Phi(O_t, q_t) = Y_t,$$

with constant residues at every node. Write $W := \Phi(B_0) \supset \{Y_t\}$: W is a closed $\mathrm{PSL}_2(\mathbb{R})$-invariant algebraic subset of $\mathcal{M}_{g_i,n_i}$ containing the non-constant holomorphic disk $\{Y_t\}$ (regardless of its dimension, which is 1 when Theorem 4.3(2) applies, see Appendix A for the exceptional factors).

Regenerate: apply Proposition 4.9 (MD plumbing with constant residues) to the section through (O_{t_*}, q_{t_*}): there is a holomorphic map

$$H : \mathbb{D} \times \mathbb{D} \longrightarrow \overline{\mathcal{Q}}_g, \qquad H(\varepsilon, s) \in \Omega V_0 \cup \partial, \quad H(\varepsilon, s) \in \Omega V_0 \ (\varepsilon \neq 0),$$

with $H(0,s)$ the boundary family (O_{t_*+s}, q_{t_*+s}) and $\Phi(H(\varepsilon,s))$-projection a holomorphic 2-parameter family of curves in $\overline{\mathcal{M}}_g$ with $\Phi(H(\varepsilon,s)) \in V_0$ for $\varepsilon \neq 0$ — property (3) of Proposition 4.9, applied with $\widetilde{M} = \Omega V_0$ and with the boundary affine manifold $B_0 \supset \{(O_t, q_t)\}$ of Lemma 5.6; its hypotheses are met: the residues of the section are constant by Proposition 5.8, and every point of B_0 is DM-reachable from ΩV_0 by the very definition of B_0 (Lemma 5.6).

Lemma 6.1 (Component identification)**.** *Let $\widetilde{X}_{0,s}$ be the stable curve underlying $H(0,s)$, $s \in \mathbb{D}$, and let $\widetilde{X}^{(i)}_{0,s}$ denote its i-th component as an element of $\mathcal{M}_{g_i,n_i}$ (well-defined: by Proposition 4.9(4), no node on the i-th component opens as s varies at $\varepsilon = 0$, so the i-th vertex of the dual graph is persistent on the disc and its component is a 1-parameter family of smooth n_i-pointed curves). Then*

$$\widetilde{X}^{(i)}_{0,s} \;=\; Y_{t_*+s} \qquad \text{for all } s \text{ in the disc},$$

and in particular $\partial_s \widetilde{X}^{(i)}_{0,s}\big|_{s=0} = \dot{Y}_{t_*} \neq 0$.

Proof. By Proposition 4.9(2), the boundary family $s \mapsto H(0,s)$ is the given boundary disc of stable differentials, i.e. $H(0,s) = (O_{t_*+s}, q_{t_*+s})$: the underlying pointed Riemann surface of the i-th component is, by definition of the forgetful map Φ, $\Phi(O_{t_*+s}, q_{t_*+s}) = Y_{t_*+s}$ (Corollary 5.10). The projection $\widetilde{X}^{(i)}_{0,s}$ of the stable model of the total curve underlying $H(0,s)$ is normalized with the marked points at the relevant nodes; the stable model is unique (stable curves have finite automorphisms, and the nodal structure is fixed on the disc since no node opens, property (4)), so $\widetilde{X}^{(i)}_{0,s} = Y_{t_*+s}$ is the *same* family of stable pointed curves. (The only freedom would be a simultaneous automorphism, trivial for stable n_i-pointed curves with $2g_i - 2 + n_i > 0$.) Hence $\partial_s \widetilde{X}^{(i)}_{0,s}\big|_{s=0} = \frac{d}{dt} Y_t\big|_{t=t_*} = \dot{Y}_{t_*} \neq 0$ by the choice of t_*. □

Equivalently, after quotienting the $\mathbb{C}^*$-direction, there is a holomorphic map

$$h : \mathbb{D}^2 \longrightarrow \overline{\mathcal{M}}_g, \qquad h(\varepsilon, s) \in V_0 \ (\varepsilon \neq 0), \qquad \partial_s\big(\pi_i h\big)(0,0) = \dot{Y}_{t_*} \neq 0,$$

and $h(0,s)$ is the boundary stable-curve family lying over the boundary differentials (O_{t_*+s}, q_{t_*+s}) of the section.

6.2 The special fibre lies in $\overline{C_0}$: the contradiction

We resume from the holomorphic map

$$h : \mathbb{D}^2 \longrightarrow \overline{\mathcal{M}}_g, \qquad h(\varepsilon, s) \in V_0 \text{ for } \varepsilon \neq 0, \qquad \partial_s\big(\pi_i h\big)(0,0) = \dot{Y}_{t_*} \neq 0,$$

constructed above.

Lemma 6.2 (The $t = 0$ premise of hypothesis (ii) on the face $\{T_1 = 0\}$)**.** *Let U be a plumbing neighbourhood as above, T_1 the coordinate of the extra node $\mathcal{N}$, $G = \Gamma$, $G' = \Gamma \cup \mathcal{N}$. Then*

$$\overline{\overline{V_0} \cap \Delta_\Gamma} \cap U \cap \{T_1 = 0\} \;=\; \overline{V_0} \cap U \cap \{T_1 = 0\} \;\subseteq\; \overline{C_0} \cap U,$$

i.e. the premise of hypothesis (ii) holds (with the right side equal to the left), for V_0 at $t = 0$ on the face over the extra node.

Proof. The equality $\overline{\overline{V_0} \cap \Delta_\Gamma} \cap U \cap \{T_1 = 0\} = \overline{V_0} \cap U \cap \{T_1 = 0\}$ is a general fact about a complex-analytic variety $\overline{V_0}$ meeting a normal-crossings boundary divisor with local equation $T_1 = 0$: the analytic closure commutes with restriction to a divisor on an open set where the intersection is proper — here properness holds on U because $\overline{V_0} \cap \Delta_\Gamma = C_0$ is a curve and $\{T_1 = 0\}$ cuts C_0 in finitely many points, so on a small enough U the cut locus is a finite set of points, and closure of a finite-type analytic family of points is the union of the limits. It remains to show $\overline{V_0} \cap U \cap \{T_1 = 0\} \subseteq \overline{C_0}$.

For the inclusion we proceed in two steps: we first replace the face $\{T_1 = 0\}$ by the ambient closure $\overline{\Delta_\Gamma}$ (no analyticity beyond complex-analytic set theory is needed), and then exclude extra one-dimensional components inside that closure by a dimension count on ΩV_0.

Step 1: closure of the boundary piece commutes with the stratum decomposition. Shrink U so that $\overline{\Delta_\Gamma} \cap U = \Delta_\Gamma^\circ \cap U$, where $\Delta_\Gamma^\circ \supset \Delta_\Gamma$ is the union of the finitely many strata of $\overline{\mathcal{M}}_g$ whose closures contain Δ_Γ (the graphs obtained from Γ by contracting further nodes). Since Δ_Γ° is Zariski-locally closed in $\overline{\mathcal{M}}_g$ (it is open in its own closure, as is any stratum union of a normal-crossings divisor decomposition), the set $W_0 := \overline{V_0} \cap \Delta_\Gamma^\circ$ is a complex-analytic subset of a normal analytic open set (the ambient space minus finitely many strata disjoint from Δ_Γ); hence $\overline{W_0} \cap U$ is complex-analytic of pure dimension 2, and W_0 is Zariski-dense in $\overline{W_0} \cap U$. Every point $P \in \overline{V_0} \cap \overline{\Delta_\Gamma} \cap U = \overline{W_0} \cap U$ is therefore a limit of points of W_0. The curve

$$C_0 \;:=\; \overline{V_0} \cap \Delta_\Gamma \;=\; W_0 \cap \Delta_\Gamma$$

is Zariski-dense in $\overline{W_0} \cap U$: otherwise some irreducible component S of $\overline{W_0} \cap U$ would not meet Δ_Γ, but $S = \overline{S \cap W_0}$ by analyticity and $W_0 \setminus \Delta_\Gamma = \varnothing$. Consequently every point P is a limit of points of C_0:

$$\overline{V_0} \cap \overline{\Delta_\Gamma} \cap U \;=\; \overline{V_0 \cap \Delta_\Gamma} \cap U \;=\; \overline{C_0} \cap U,$$

which is the load-bearing closure identity (the middle equality of the lemma). Since $\Delta_{\Gamma \cup \mathcal{N}} \cap U \subset \overline{\Delta_\Gamma} \cap U$ (on U the strata adjacent to Δ_Γ are nested in its closure), it follows that

$$\overline{V_0} \cap U \cap \{T_1 = 0\} \;=\; \overline{V_0} \cap \Delta_{\Gamma \cup \mathcal{N}} \cap U \;\subseteq\; \overline{V_0} \cap \overline{\Delta_\Gamma} \cap U \;=\; \overline{C_0} \cap U.$$

Step 2 (only a consistency check; not needed for the conclusion): the face $\mathcal{F} := \overline{V_0} \cap U \cap \{T_1 = 0\}$ can in fact have no 1-dimensional component. Indeed, suppose it had one, $\mathcal{F}_1$. Step 1 gives $\mathcal{F}_1 \subset \overline{C_0}$, a curve, so $\mathcal{F}_1$ would be a boundary arc of $\overline{V_0}$ across the extra node accumulated to $\overline{C_0}$. Pulling back the $\mathrm{GL}_2^+(\mathbb{R})$-action on ΩV_0 through the face projection Φ, $\mathcal{F}_1$ supports a 1-parameter family of stable differentials in the boundary of ΩV_0 whose $\mathrm{GL}_2^+(\mathbb{R})$-closure $\widehat{\mathcal{F}}_1$ is an affine invariant manifold (Theorem 4.1(1)) projecting to the closed invariant algebraic set $\Phi(\widehat{\mathcal{F}}_1) \supset \mathcal{F}_1$ (Theorem 4.1(3)). Now ΩV_0 has complex dimension 3 with $\dim_{\mathbb{C}} \Phi(\Omega V_0) = 2$ (Theorem 4.3(1)), so the generic fibre of $\Phi|_{\Omega V_0}$ — which contains the $\mathbb{C}^*$-orbit and, over an interior point, equals the complex 1-dimensional space of disc-generating differentials — has complex dimension 1. Passing to the boundary face over $\mathcal{F}_1$, the fibre can only be smaller: a stable curve on a boundary divisor has finite automorphism group, hence the stable differentials over a fixed point of $\mathcal{F}_1$ in the boundary of ΩV_0 form a discrete union of $\mathbb{C}^*$-orbits, and

$$\dim_{\mathbb{C}} \big(\Phi|_{\widehat{\mathcal{F}}_1}\big)^{-1}(x) \;\le\; 1 \qquad (x \in \mathcal{F}_1 \text{ generic}).$$

As $\dim_{\mathbb{C}} \widehat{\mathcal{F}}_1 \le \dim_{\mathbb{C}} \Omega V_0 - 1 = 2$ (it lies in the boundary of the 3-fold), this forces $\dim_{\mathbb{C}} \Phi(\widehat{\mathcal{F}}_1) \le 1$: the projection of the invariant closure is a curve, so $\mathcal{F}_1$ — being a 1-dimensional irreducible component of that curve — would have to *be* $\Phi(\widehat{\mathcal{F}}_1)$ itself, an invariant curve. But Theorem 4.7(1)-(4) identifies the boundary of ΩV_0 over the face through $\overline{C_0}$ with the central Bers set of the boundary piece: its projection lies over the face of C_0, i.e. over the discrete set $\overline{C_0} \cap \{T_1 = 0\}$ (stable-model uniqueness: the smoothing direction tangent to $\overline{V_0}$ at a face point regenerates along C_0). A 1-dimensional $\Phi(\widehat{\mathcal{F}}_1)$ cannot lie over a discrete set, contradiction. Hence $\mathcal{F}$ is 0-dimensional, and Step 1's inclusion is an equality with a finite set. (This strengthens the inclusion but is not used in Section 6, which only needs $\overline{V_0} \cap U \cap \{T_1 = 0\} \subseteq \overline{C_0}$.) □

Proposition 6.3 (Contradiction)**.** *Under the standing assumptions, the existence of such an h is impossible. Consequently the non-constant alternative cannot occur, and $t \mapsto Y_t$ is constant for $|t| \ll 1$.*

Proof. By property (4) of Proposition 4.9, the plumbing coordinate T_1 of the collapsed node $\mathcal{N}$ vanishes identically on $h(\{0\} \times \mathbb{D})$ but is nonzero on $h(\varepsilon, s)$ for $\varepsilon \neq 0$; hence the holomorphic map $h_0 := h(0, \cdot) : \mathbb{D} \to \overline{\mathcal{M}}_g$ satisfies:

(a) $h_0(s) \in \{T_1 = 0\} \cap \overline{V_0}$ for all s: indeed $h_0(s) = \lim_{\varepsilon \to 0} h(\varepsilon, s)$ with $h(\varepsilon, s) \in V_0$ (closedness of limits), and $T_1(h_0(s)) = \lim T_1(h(\varepsilon, s)) = 0$;

(b) h_0 is non-constant, with $\partial_s(\pi_i \circ h_0)(0) = \dot{Y}_{t_*} \neq 0$.

Let U be the plumbing neighbourhood of $h_0(0)$ on which the coordinates $(T_1, \dots, T_N; m)$ of Section 4 are defined; after shrinking $\mathbb{D}$, $h_0(\mathbb{D}) \subset U$. Hypothesis (ii) applied with $G = \Gamma$, $G' = \Gamma \cup \mathcal{N}$ (reading $\Delta_{G'} \cap U = \Delta_\Gamma \cap \{T_1 = 0\} \cap U$ in the plumbing model: the intersections living on U of the two adjacent strata differ exactly by $T_1 = 0$) gives

$$\overline{V_0} \cap U \cap \{T_1 = 0\} \;=\; \overline{\overline{V_0} \cap \Delta_\Gamma} \cap \{T_1 = 0\} \cap U \;=\; \overline{C_0} \cap \{T_1 = 0\} \cap U \;\subseteq\; \overline{C_0} \cap U.$$

This identification at $t = 0$ is the content of Lemma 6.2: the first equality $\overline{\overline{V_0} \cap \Delta_\Gamma} \cap \{T_1 = 0\} \cap U = \overline{V_0} \cap \{T_1 = 0\} \cap U$ is a general analytic fact about closure commuting with a proper divisor cut, and the inclusion $\subseteq \overline{C_0}$ is proved there from the totally geodesic structure of V_0 via Theorem 4.1(3) and Theorem 4.7(1). Therefore

$$h_0(\mathbb{D}) \;\subset\; \overline{V_0} \cap \{T_1 = 0\} \cap U \;\subseteq\; \overline{C_0} \cap U.$$

Now $\overline{C_0} \subset \Delta_\Gamma$ is a curve in the product $\prod_j \mathcal{M}_{g_j,n_j}$ with $\pi_i(C_0) = \{Y_0\}$; the closure $\overline{C_0}$ still projects under $\hat{\pi}_i$ to $\{Y_0\}$ (Lemma 6.4 below, noting that inside U the i-th component is unbroken, so the projection extends continuously as $\hat{\pi}_i$ to $\overline{C_0} \cap U$). Hence $\hat{\pi}_i \circ h_0$ is a continuous, holomorphic map to the single point $Y_{t_*} = \hat{\pi}_i(h_0(0))$, i.e. constant; but its derivative at $s = 0$ is $\partial_s(\hat{\pi}_i \circ h_0)(0) = \partial_s(\pi_i h)(0,0) \neq 0$, a contradiction. □

Lemma 6.4 (Constant projection on the closure)**.** *Let $C \subset \Delta_\Gamma$ be a closed analytic curve with $\pi_i(C) = \{Y_0\}$, and let $\overline{C}$ be its closure in $\overline{\mathcal{M}}_g$. Then $\pi_i(\overline{C} \cap \Delta_\Gamma) = \{Y_0\}$; more precisely, if $\hat{\pi}_i$ denotes the extension of π_i to a neighbourhood of $\overline{C}$ in which the i-th component of the stable curves remains a single component (a plumbing neighbourhood), then $\hat{\pi}_i(\overline{C}) = \{Y_0\}$.*

Proof. π_i is the projection of a product, hence continuous and closed in a neighbourhood of $\overline{C} \cap \Delta_\Gamma$; a continuous map constant on a set is constant on its closure in the domain. □

6.3 Globalization

The local argument of the preceding sections (Sections 5 through the contradiction Proposition 6.3) shows: after possibly shrinking Δ around 0, the holomorphic map $t \mapsto Y_t$ of Lemma 5.3 cannot be non-constant; hence there is a neighbourhood $\Delta' \ni 0$ with $Y_t = Y_0$ for all $t \in \Delta'$. (The argument is anchored at $t = 0$ because it uses the totally geodesic structure of V_0; this is essential and corrects a previous version which instead invoked a false re-centring of the whole argument at arbitrary t_0 — V_{t_0} is not totally geodesic for $t_0 \neq 0$.) Now $t \mapsto Y_t$ is a holomorphic map from the connected disc Δ to $\mathcal{M}_{g_i,n_i}$ (Lemma 5.3), and it agrees with the constant map $t \equiv Y_0$ on the open nonempty set Δ'. By the identity theorem for holomorphic maps between complex manifolds (target Hausdorff), $Y_t = Y_0$ for all $t \in \Delta$. This proves Theorem 3.1. □

6.4 Where each hypothesis is used

- V_0 totally geodesic and 2-dimensional: Theorem 4.3 applies to V_0, making ΩV_0 an affine-invariant 3-fold (this enters the dimension dichotomy of Lemma 5.6 through $(O_0, q_0) \in \partial\Omega V_0$), and gives the no-poles/cylinder-only degeneration of Theorem 4.3(3) feeding Theorem 4.7(4).
- $\dim \partial V_t = 1$: makes ∂V_t a curve so that the plumbing slope analysis of Lemma 5.1 applies and so that the degeneration data are well-defined stable differentials (the limit along a 1-dimensional boundary curve).
- Hypothesis (i): smooth triviality gives the limiting $\lim_{t\to 0} y_t$, the analytic total space of Lemma 5.3, and the closed-analytic incidence level sets used in Lemma 5.6 and Proposition 5.8.
- Hypothesis (ii): compatibility in deeper strata is used crucially in Lemma 5.1 (the collapsed-node data and slopes at $t \neq 0$ equal those at $t = 0$, computed in the deeper stratum for V_0) and in the contradiction step (Proposition 6.3, where $\overline{V_0} \cap U \cap \{T_1 = 0\} \subseteq \overline{C_0}$ is exactly hypothesis (ii)) and the globalization step.
- $\pi_i(\partial V_t) = \{\text{pt}\}$: produces the single-valued map $t \mapsto Y_t$, and at $t = 0$ gives $\pi_i(\overline{C_0}) = \{Y_0\}$ used in Proposition 6.3 via Lemma 6.4.

A Low-complexity factor cases

The main argument (Sections 5–6) used the dynamics of $\mathrm{PSL}_2(\mathbb{R})$ on strata and the exclusion of closed totally geodesic surfaces (Theorem 4.3) in the factor $\mathcal{M}_{g_i,n_i}$. Here we record the remaining

low-complexity factors, where the same conclusion holds either vacuously or by a mild adaptation. There is nothing to change in Lemma 5.1, Lemma 5.3, Proposition 5.8 or the globalization: they use only the plumbing coordinates, the hypotheses (i)–(ii), and Rado/Bishop — never the $\mathrm{PSL}_2(\mathbb{R})$-structure.

Factor a point: $(g_i, n_i) = (0, 3)$**.**
$\mathcal{M}_{g_i,n_i}$ is a single point. Then $\pi_i(\partial V_t)$ is automatically the single point; $Y_t = Y_0$ and there is nothing to prove.

One-dimensional factor: $\dim_{\mathbb{C}} \mathcal{M}_{g_i,n_i} = 1$ **(i.e.** $(g_i, n_i) = (0, 4)$**, or** $(1, 1)$ **as an orbifold:** $\mathcal{M}_{1,1}$ **is a 1-dimensional orbifold curve, not a point).**
Let $\{T_1 = 0\}$ be a boundary divisor meeting C_0 (Lemma 5.1); by hypothesis (ii) (its $t = 0$ premise being Lemma 6.2), $\overline{V_t} \cap \{T_1 = 0\} \cap U = \overline{C_t} \cap U$ for all t, and this cut is a 0-dimensional analytic cycle Z_t (a proper intersection of a surface with a divisor), varying holomorphically in t by hypothesis (i). Hence $\pi_i(Z_t) \subset \mathcal{M}_{g_i,n_i}$ is a holomorphically varying 0-cycle; on the other hand $Z_t \subset \overline{C_t}$, and $\pi_i(\overline{C_t})$ is the singleton $\{Y_t\}$ (Lemma 6.4), so $\pi_i(Z_t) = \{Y_t\}^{\oplus \deg Z_t}$ as a 0-cycle concentrated at Y_t. The support $t \mapsto Y_t$ of a holomorphic family of 0-cycles of constant degree concentrated at a point is a holomorphic map (it is the support of the image cycle); by Lemma 5.3 it is the map already denoted $t \mapsto Y_t$. Since Z_t is a flat deformation of Z_0, which is supported at Y_0, and a flat family of 0-cycles of constant degree with singleton support has constant support (the cycle map is continuous and the support of Z_0 is Y_0), we get $Y_t = Y_0$ for all small t, hence for all t by the identity theorem. (This argument does not use any $\mathrm{PSL}_2(\mathbb{R})$-structure, so it applies to every factor of dimension 1, including $(0, 4)$ and $(1, 1)$.)

Genus zero with $n_i \geq 5$**:** $(g_i, n_i) = (0, n_i)$**,** $n_i \geq 5$**.**
The argument proceeds verbatim as in the main case. The strata of meromorphic quadratic differentials on $\mathbb{P}^1$ still carry the complex-affine period-coordinate structure, the $\mathrm{GL}_2^+(\mathbb{R})$-action, and the plumbing/regeneration theory of Section 4 (the references cited there remain valid in genus zero), and the membership, regeneration, and contradiction arguments of Sections 5–6 apply verbatim. The exclusion of a 2-dimensional Φ-image is not needed here: Proposition 6.3 is g_i-independent, using only plumbing coordinates and hypothesis (ii). Hence $Y_t = Y_0$.

Genus two: $g_i = 2$**,** $n_i \geq 0$**.**
$\mathcal{M}_2$ contains McMullen's Hilbert modular surfaces [5]: genuine totally geodesic surfaces, projections of affine invariant 3-folds with 2-dimensional base. If $\Phi(B_0)$ were such a surface $W_H \subset \mathcal{M}_{2,n_i}$, then the boundary face of ΩV_0 over the corresponding face of $\partial\mathcal{M}_g$ would dominate W_H; but ΩV_0 fibres over $V_0 \subset \mathcal{M}_g$ with 1-dimensional fibres away from the boundary, and taking closures identifies the fibre of $\Phi|_{B_0}$ generically with the disc bundle of the boundary stratum: a 2-dimensional $\Phi(B_0)$ is possible, and in that case $\{(O_t, q_t)\}$ varies in a closed totally geodesic surface. (Here Input (Mem) is still assumed, as in Section 5.) However, hypothesis (ii) and the regeneration argument of Section 6 apply literally: the conclusions of Proposition 6.3 (which uses only plumbing coordinates and hypothesis (ii), not the value of g_i) force $Y_t = Y_0$ in this case as well. (The only place in Sections 5–6 where g_i enters is through Theorem 4.3(2), used to exclude a priori the possibility that $\Phi(B_0)$ is a totally geodesic surface; replacing this exclusion requires no further argument, since the contradiction of Proposition 6.3 is deduced independently of the nature of $\Phi(B_0)$.)

B Status of the main statement and the single external input

For the convenience of the reader we collect the logical structure. Theorem 3.1 is proved from: (i) the hypotheses of the problem statement; (ii) the four quoted inputs (Theorems 4.1, 4.3, 4.7 and Proposition 4.9); and (iii) the single *additional analytic input*, Input (Mem) (p. 10), whose content is: the limiting stable differentials of the whole deformation remain in the affine manifold generated from the V_0 boundary piece. Input (Mem) is expected to follow from the full strength of the boundary theory of affine invariant manifolds applied to the total space of the family $\mathcal{V}$; isolating it here makes the logical dependencies explicit and confines the non-elementary analytic content of the argument to a single, precisely stated place.

Consistency certificate with the problem statement. The conclusion of Theorem 3.1 is exactly the conclusion asked for in the problem: $\pi_i(\partial V_t) = \{Y_0\}$ literally for *all* $t \in \Delta$, not merely for small t (the identity theorem of Section 6 lifts the local constancy near $t = 0$ to the whole connected disc). The hypotheses of Theorem 3.1 match the problem statement term-for-term; the only logical residue relative to the problem's literal request is the single unproved analytic input Input (Mem), isolated and labelled — not concealed.

Remark B.1 (Self-diagnosis: where the missing anchor sits). At the level of stable curves, the missing anchor of the argument is exactly the $t \neq 0$ form of the premise of hypothesis (ii) — Lemma 6.2 supplies that premise only at $t = 0$, which is why the contradiction of Proposition 6.3 is staged at $t = 0$ via regeneration into V_0. Input (Mem) sits precisely one Φ-level above this: it is the same compatibility, lifted from stable curves to the quadratic differentials and their affine-linear invariance equations.

References

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